



Imagining Public Space in Smart Cities:

a Visual Inquiry on the Quayside Project by Sidewalk Toronto

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Abstract

Recently, the 'Smart City' label has emerged as a popular umbrella term for numerous projects around the world that claim to offer an enhanced urban experience, often provided in collaboration with international companies through private-public partnerships. As smart cities pledge to create long-term economic sustainability and progressive form of urban entrepreneurialism, it is getting important to highlight risks such as the reduced role of the public sector, technological dominance and data privacy. In contrast to more a conventional, long-term, holistic master planning, a technologically pre-determined form of Smart City endangers the emancipator usage of public spaces as spaces of diversity, creativity, inclusive citizen participation and urban sustainability.

This research approaches the concept of Smart Cities as a future category and, thus, targets to develop a comprehensive visual analysis based on architectural representations in the form of computer-generated images (CGI's). The Quayside project, a notable and widely criticized urban development project, by Sidewalk Toronto, a cooperation between Waterfront Toronto and Sidewalk Labs which is a sister subsidiary of Alphabet Inc., has been selected as Smart City case study as. Visual analysis was conducted by using the theoretical frame advocating 'Coordinating Smart Cities' in contrast to 'Prescriptive Smart Cities' by Richard Sennett. In addition to Sennett's concept of 'Incomplete Form', Jan Gehl's 'Twelve Quality Criteria' was used as coding categories to elaborate the content analysis which was followed by semiological and compositional interpretations. Visuals have been investigated in three sequential sets and analyzed focusing on time-based comparative frequency counts for sets of visuals. Concentrating on how future public spaces are illustrated, the study aims to uncover and to discuss how Smart Cities are being imagined and advertised.

Keywords

Smart City, Smart Urbanism, Neoliberal Planning, Public Space, Local Governance, Digitalization, Visual Theory, Architectural Representations, Sidewalk Toronto, Quayside Project

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Introduction

'Smart City' has become a discourse that dominates the discussions where the future of urban life is debated and imagined. Evidently, it is a fruitful term to use through neoliberal 'inter-urban competition' to promote cities. It is most alluring for local governances since Smart Cities are contributing remarkably to the overall competitiveness of the city with promises of better predictability and efficiency. The term is equally attractive for private companies since technologically driven cities are enabling an immense global market for all sorts of new products aspiring a more secure, connected and therefore optimized urban experience. In contrast, Smart Cities incorporate high risks of widened social inequity, techno-managerialism, data privacy abuse and overstimulation. David Harvey (1989) argues about that this new "urban entrepreneurialism" is depending on public-private partnerships while "focusing on investment and economic development with the speculative construction of place rather than amelioration of conditions within a particular territory as it's immediate (though by no means exclusive) political and economic goal."

The aim of the research is to reveal how everyday life is being imagined, designed or speculated in Smart Cities, holding the promise of technological advancements with the possibility to acquire prospective places that are more inclusive, creative and sustainable. My approach is focusing on the production of everyday life in public spaces. The projected framework is a critical tool to evaluate urban visions while keeping in mind the urban environments of tomorrow will potentially be data-driven, emancipating, accessible by all, sustainable and sensitive to human rights and privacy. Thus, the aim of the thesis is to create a framework to discuss emerging public spaces in Smart Cities and how they are imagined to be. Even though we have started to experience smart applications in cities, the smart urbanism implies the digital transition, a growing narrative about the close future. Thus, the critical framework of this study will inclusively unfold itself on evaluating computer-generated images (CGI's) from one of the leading Smart City projects – Sidewalk Toronto – through critical visual analysis using various methodologies in combination. Therefore, it is a visual inquiry on how public space is being imagined in Smart Cities, discussing the possible courses that may possibly be contributing to such a quest in the future.

Since the turn of the 21st century, the everyday life became a focal point of interest in urban theory. This interest is due to the practical focus of human factor or human-centric approach but also, taking into account the interaction of humans at a wider range, it is related to the built environment as non-human factors, following actor-Network Theory (ANT) perspectives (Latour, 2005). Furthermore, according to Baxter (2017), there is a growing interest in architecture and built environment in new urban scholarship. My approach is evidently centralizing the product or the medium of architecture with CGI's as the outputs of urban imagination. What is more, I propose to look at them as images through a critical lens that encompasses different sites and modalities for visual interpretations (Rose, 2016) as a network of production. Baxter argues that understanding CGI's as the network is a crucial process that hasn't yet provoked much attention in the urban theory. However, since focusing into the drawings or visuals involve "the amount of work, interactions and technologies involved in the regeneration of cities and implicitly in the production of inequality" (Baxter, 2017), it is worthwhile to use them as a standing point. Accordingly, Rose et al. (2014) argue that CGI's attract "relatively little attention as a new form of visualizing the urban", claiming that they are not only "persuasive marketing images", but they also function as "problematic interfacial sites created by many different kinds of work". Along these lines, my thesis is fulfilling an emergent zone in Urban Studies dealing with architecture through its network of production representation and impacts on the future urban environments

Research Questions

- What is a smart public space? What are the spatial conditions, functions, processes or materialities that stimulate the smart public space?
- How do we interact with/into the public space (indoors-outdoors, virtual-physical) in the era of artificial revolution? How are artificial intelligence (AI) and Information and Communication Technologies (ICT's) affecting the formation/production of public spaces?
- How is public space represented in Smart City projects?

The Structure of the Study

The study is designed to have a clear structure to provide an in-depth discussion of how public spaces in Smart Cities are represented. As mentioned before the research is based on a visual inquiry of the Quayside project by Sidewalk Toronto. Due to the visually oriented nature of the study, a coherent explication of the methods that will be used is holding a crucial place. At the methodology chapter, the general approach will briefly overview how the analysis will be done in terms of visual culture and theory. This part will be followed by the presentation of the chosen visual methodologies in relation to the targeted analysis, finalized with a discussion on the limitations of the research, in particular.

The next chapter will be about theoretical backgrounds presenting selected concepts and ideas on design and imagination of public spaces. I am aiming to posit my standing point within the production of urban imaginaries in the form of CGI's. Therefore, the theoretical part will start a critical inquiry of how architectural/urban design is being produced, followed by a discussion about the role of representation within the design processes though recent cases on ethnographic research. Having started with how the urban environment is conceived, the theoretical part will continue briefly with how public space has evolved from pre-industrial to post-industrial considering the emergence of the digital revolution. Next, I will focus on the notion of understanding what Smart Cities are and how they function with a review of smart applications, the digital transformation of the urban with a critical point of view including NGO declarations and manifestos. I will conclude with the introduction of the overarching frame of prescriptive and coordinating Smart Cities that I borrowed from Richard Sennett's book titled *Building and dwelling: Ethics for the City* (2018) and finally, I will present categories from Sennett and Gehl that I will use to code CGI's.

What comes next is the presentation of the chosen case study; Quayside project by Sidewalk Toronto. I will explain why I chose this project with it's brief history and present the current state of the project. This part will include the visions and reflections of designers and related agencies or institutions on the project, followed by related smart applications/initiatives and the criticism available from newspapers, academic articles and public debates. After introducing the case, I will unfold the visual analysis with findings, as the result of the attained visual inquiry with several representative pictures from visual datasets. Finally, I will discuss my analysis with the overarching theoretical frame. This part will be a concise summary of the study providing my interpretations and concluding remarks.

Methodology

General Approach

The research is designated to set a critical frame that questions the relationship between society and environment whether it is natural, constructed, imaginary or virtual. Thus, it will mainly use critical urban theory and reflexivity as theoretical backgrounds. The aim of the research is to contribute with a creative and stimulating output for an 'incomplete form' for public space that Smart Cities of the upcoming future. Accordingly, the main field of this research is to focus on how Smart Cities, particularly the public space in that future built environments, are being envisioned. This attempt encompasses the everyday life of Smart Cities whether they are presumed or imagined. With this aim, the dominant methodology will be the structured examination of the visuals that are digitally produced, in other words CGI's (computer-generated-images). This spotlight requires a detailed scheme of appropriate methodologies to evaluate digital visualizations as interpretations of an imagined yet aimed-to-be persuasive future urban life.

As the case study of visual inquiry, I have chosen the Quayside urban development project by Sidewalk Toronto which is a cooperation between Waterfront Toronto and Sidewalk Labs. The project is one of, potentially the most, debated Smart City projects which has drawn attention from both academic circles and the general public. The research would principally use the 'case study' approach as a qualitative method; studying in-depth analysis and eventually visual evaluation of future public spaces in Sidewalk Toronto. Bloor & Wood defines the method case study (2006) as an "exploration of a 'bounded system'", a case may be many different things; from an individual to a city or an institution. Thus, case studies are used centrally in the vast field of social sciences as a method largely unstandardized yet used to "supplement other research methods including quantitative techniques" (Bloor & Wood, 2006). How to choose a case is essential, by definition, as put by Hammond and Wellington (2013) "a case is literally an example of something – a unit of analysis". The selection of case or cases must be representative of a particular topic of interest in relation to a problem. Bloor & Wood point out that the accessibility of a case may affect such choice. The case that will be presented is benefitting multiple ethnographic data collections, such as brief descriptions and status, the vision of the architects and/or the developers, smart applications and initiatives related to the projects, criticism available through newspapers and academic articles as well as public debates and interviews.

The principal method to be used is visual, mainly based on Gillian Rose's (2016) approach through her book titled *Visual Methodologies*. A convenient mix of visual methods will be defined and developed through the main framework of the author and other related authors on the topic. Second, the focal point will move to architectural representations and drawings. An intensive discussion will be conducted on the notion of drawings of multiple kinds for the practice of architecture and urban design. By doing so the representative CGI's, as produced visuals, will be situated adequately to the realm of visual culture and reproduction. The findings from the visual evaluation of the selected images will be explained in the following chapters.

Due to the intersectional disposition of the research, I propose to start with a 'purposive literature review', which will encompass the field where the public space is addressed spatially and possibly through digitalization. Hammond and Wellington (2013) suggest that a purposive review would handle the literature critically while "staying sensitive" to the context of the sources. This process extends further to review the co-evolution of public life and public space through examples showing how smart applications are being used in the public realm.

The methodological explanation or the design of the research is particularly important since it is dealing exclusively with the visuals – through an unavoidably subjective manner. Consequently, in the methodology part, I will lean towards a peculiar framework to position my starting point at visual culture considering architectural/urban representations and the use of public space of which I will elaborate in later chapters to relate my chosen methods to the theoretical frame.

The Critical Framework: The Visual

“With the word “picture,” we think first of all of a copy of something. Accordingly, the world picture would be a painting, so to speak, of what is as a whole. But “world picture” means more than this. We mean by it the world itself, the world as such, what is, in its entirety, just as it is normative and binding for us.”

(Heidegger, 1977, s. 129)

The everyday life we experience in contemporary settings is dominated by the image, and this is a global fact. At large it is a part of the consequences of neoliberal globalism and the infamous culture of consumption, but visual hegemony is also closely related to the widely available and emergent technologies, regarded in general as a Western phenomenon. For instance Rose (2016) puts this as follows: “the visual is central to the cultural construction of social life in contemporary Western societies”. Mirzoeff (1998) is highlighting “the emergence of visual culture as a subject has contested this hegemony” of Western conceptualization of spoken word as a higher category compared to visual representation through the course of history. Pallasmaa (2005) emphasizes that “sight has historically been regarded as the noblest of the senses, and thinking itself thought of in terms of seeing.” Berger (1972) points out the ‘obsession of property’ as a constitutive characteristic of Western society and culture as a whole, a “false rationalized image of itself”.

The world, not exclusively the West or Global North, is even more visual than how it used to be; it is an ‘ocularcentric’ world that is governed by a ‘scopic regime’. Mirzoeff (1998) identifies the current exposure to the ‘cultural turn’ that is fundamental at larger scales of human and social sciences where “meaning is thought to be produced- constructed - rather than simply ‘found’”. Such constructed social categories do take dominant visual forms as well, whereas “the modern connection between seeing and knowledge is stretched to the breaking point in postmodernity” (Rose, 2016). Mitchell (1995), coins the term as the ‘pictorial turn’ that triggers ‘peculiar friction and discomfort’ at the wide range of intellectual research, representing a fear from a technically enabled global danger of “a culture (that is) totally dominated by images”.

Digital imaging technology is an emerging ‘currency’ that is “disrupting familiar practices of image production” (Mitchell W. J., 1992). Accordingly, Mirzoeff (1998) asserts that the visual culture, emphasized by the cultural turn, has not solely been based on pictures but tends to visualize its existence insistently. That is radically different from the ancient perception of the world that once understood as a book, as he claims. In that world, or what is explicitly implied as the non-Western world, “pictures were seen not as representations, artificial constructs seeking to imitate an object, but as being closely related, or even identical, to that object.” Mirzoeff continuously explains how the perceived reality has shifted throughout the course of modern history:

"Perspective images sought to make the world comprehensible to the powerful figure who stood at the single point from which they were drawn. Photographs offered a potentially more democratic visual map of the world. Now the filmed or photographic image no longer indexes reality because everyone knows it can be undetectably manipulated by computers. The paradoxical virtual image 'emerges when the real-time image dominates the thing represented, real-time subsequently prevailing over real space, virtuality dominating actuality and turning the concept of reality on its head' (Virilio, 1994, s. 93)."

Furthermore, we are increasingly experiencing an overexposure of information, images and screens consolidated with ubiquitous smart technologies as part of the 'digital turn'. Thus, the digital transformation is exponentially affecting everyday lives. The world is 'known' digitally, think of the extensive use of the satellite imageries, augmented and virtual realities and their wide range of applications, or military purposes of image making for instance in the form of 'authoritarian visuality' (Mirzoeff, 2011).

It is central how it is offered to see the world or the image itself but it is also equally important to reflect upon who is looking at those pictures, namely the positioning of the audience or audiencing. Berger (1972) argues that "we never look just at one thing; we are always looking at the relation between things and ourselves". Visual images 'work like languages', they are a kind of language as for instance the music, body language and facial expressions (Hall, 1997). The cultural meanings are derived from the shared understanding to translate, to interpret feelings, concepts or ideas through these mediums of languages. Nevertheless, this transportation is by no means flawless, there is always something 'lost in translation'. Hall (1997) concludes this loss strikingly as following: "Meaning is a dialogue - always only partially understood, always an unequal exchange."

At this point, Gillian Rose proposes a critical visual methodology of how to look at pictures. Those possible 'ways of seeing' will be my framework to evaluate or to understand and interpret the CGI's as Smart City imaginaries. Rose's methodology is consisting of an overarching frame containing sites and different aspects of those sites as modalities. Forming her main critical approach, Rose is strictly recommending a couple of points that are crucial for the adequacy of the visual process. First of all her framework for the visual methodology, which will shape my framework as well, "takes images seriously" and suggest a detailed look at the images without neglecting the fact that "they are not entirely reducible to their context" but holding they "own effects" as the image itself. Second, while performing a cultural practice, she insists on reflecting upon the social consequence of the visual representations with a critical account. Finally, she advocates a personal 'way of seeing' for the examiner considering about the way anyone sees is "historically, geographically, culturally and socially specific, then how you or I look is not natural or innocent". By doing so, Rose introduces us to a very important concept of such critical inquiry; the reflexivity. Rose's approach is followed by a kind of guide to looking at the pictures that are four sites of inspection where "the meanings of an image are made" ¹:

- the site(s) of the production of an image
- the site of the image itself
- the site(s) of its circulation
- the site(s) where it is seen by various audiences.

¹ The number of the sites were three at the first edition of her book. But due to the currency of evolving digital techniques she admits an addition to the sites by including the fourth site as the site of the circulation of an image through various media.

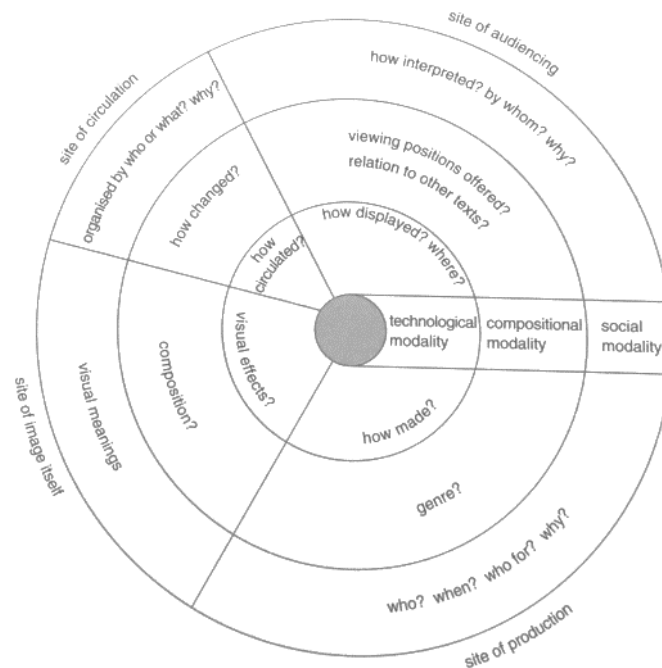


Figure 1 The sites and modalities for interpreting visual materials

In addition to the sites, she highlights three particular modalities that shape the investigated sites; technological, compositional and social. Technological modality refers to the visual technology which defines how the image has been produced and delivered. Second, the compositional modality is focused on formal strategies such as content, colour and spatial organization. Finally, the social modality is concerned about how an image is surrounded for instance with various economic, social and political relations, institutions and practices.

Thus, Rose (2016) draws the framework of critical visual methodology and furthermore, she presents and discusses various methods as ways of debating in visual culture as following; compositional interpretation, content analysis, semiology, psychoanalysis, discourse analysis, digital and audience studies. Nevertheless, she notes that one should choose and decide which sites and modalities are appropriate for the particular debate on visuals considering to use more than one method to be more inclusive since each method “have their own analytical assumptions and thus their own empirical focus”.

I feel to the urge to contemplate a bit more about the aspect of reflexivity here since Rose stresses its importance. Bloor & Wood (2006) define reflexivity as an “awareness of the self in the situation of action and of the role of the self in constructing that situation”, which associated essentially with the ‘crisis of legitimation’ in the social sciences. Accordingly, having discussed this aspect, they noticed that the academic scholarship may “abandon the pretence that the author is absent from the text” hence be more self-aware of his/her subjectivity and open to possible discussion. The aim of my research to interpret and discuss how the Smart Cities of upcoming futures are being imagined visually from today’s mindset. I argue that how we imagine a future is the commencement of its construction. Therefore my inquiry is a critical one, but still, even though structured strictly, a personal interpretation of a visionary. Keeping in mind the dimension of reflectivity and the overall aim of the research, I chose to use multiple visual methods to evaluate chosen digital visualizations as following; compositional interpretation, content analysis and semiology, in addition to some techniques from digital studies. I will explain each method and how I use them in the next chapters.

Various Methods on Visual Inquiry

1. Content Analysis

The content analysis supplies an analytical methodology borrowed from quantitative methodologies of natural sciences. According to Rose this method particularly concerned with compositional modality neglecting other sites of 'meaning-making'. The solid aim of this method is to provide "replicable and valid results" as it is in the natural sciences by basically "counting the frequency of certain visual elements in a clearly defined sample of images, and then analyzing those frequencies". Rose cites Lutz and Collins (1993) who are defending this method as a reasonable subsidiary to other qualitative methods as a "discovery of patterns that are too subtle to be visible on casual inspection". To achieve so, Rose defines step-by-step a systematical methodology.

The content analysis method starts with the finding of images that are relevant to the research question considering the representativeness of the available images. Rose lists some sampling strategies to compose meaningful datasets such as; random, stratified systematic and cluster strategies. Nevertheless, Rose (2016) asserts that sample size is affiliated with the "amount of variation among all the relevant images". Thus, even one sample will be enough to describe a typical situation if there is no variation. In such a manner I will use the stratified sampling strategy, identifying digitally available CGI's representing the public spaces at future Smart Cities as the main dataset and consequently defining subgroups as imageries associated with the selected project of interest. In my case study, the datasets are representative yet narrowed down by what is available from the official project website. The Quayside Project, has image sets are stored systematically in chronological order at the official website under the name of documents². I will use three main sets of visuals representing different phases of the projects. Those images are, circulated widely online, complete sets showing the process of image-making as well as having a high-definition resolution. The aspect of the resolution is also crucial at finding the images, considering the aim of the research is to look in detail to the selected pictures that have enough quality to enable the query. Another aspect is related to online image search through Google search engine. Online searches are open to the automated bias of result listing due to the current location, IP address and previous searches. In order to eliminate this and to sort out the digital imageries in a more analytical way, Rogers (2013) suggests to use a 'clean' version of Google which works more objectively if the searcher logs out from his/her accounts and clear all the caches and cookies, hence detaches its personal involvement from the search as much as possible. Thinking within the norms of reflexivity it is a "clean slate, free of cookies and other engine entanglements such as history and preferences". Rogers also provides other methods of tracking results by going to official websites as well as 'reverse image search' by uploading an image to locate the emergence and spread of it. Having a single and official source of images I cover such automated bias.

The second step in content analysis is to set coding categories. Rose explains this step as "attaching a set of descriptive labels (or 'categories') to the images". In order to render the method as analytical as possible, those categories should be descriptive or interpretive but should function as objective as possible, matching only with "what is 'really' there in the text or image". According to Rose, the coding should include three main characteristics as following: exhaustive, every aspect of the research theme must be covered; exclusive, not overlapping; and enlightening that is coherent.

My research depends highly on evaluating the images on how the public space in Smart Cities is being imagined based on several theoretical propositions. Rose advises using the research questions of the

² <https://sidewalktoronto.ca/documents/>

study in order to keep codes relevant, meaningful and valid after all. Thus, the codes will be central to my research, encompassing practice-based and visionary outcomes of urban thinkers of which I will give detailed explanation in the theoretical chapters. I have two mainframes to touch upon briefly in here. First, I will introduce the overarching frame that belongs to Richard Sennett (*The Public Realm*, 2017) through principles aiming an incomplete form of urban space, concerned with resilience, porosity, spectator architecture and cosmopolitanism. Then, I will complement 'Incomplete Form' categories with a second frame from Jan Gehl (*Cities for people*, 2010) setting forth 'Twelve Quality Criteria' for public space occupied with themes of protection, comfort and delight.

Besides the critical and specific character of the coding categories, Rose proposes to revise and test them as long as they are exhaustive and exclusive enough according to the purpose of the research. Then lastly concludes the content analysis by analyzing the results by counting them "in order to produce a quantitative account of their content". Rose suggests at this point 'frequency count' that may be absolute or relative, to compare with another value, i.e. change in the frequency due time or location. She warns against the difficulties of interpreting the numbers counted. For instance, she draws attention to the invisibility of some categories; "certain representations of what is visible depend on other things being constructed as their invisible opposite ". What is more, she points out the content analysis method "does not discriminate between occurrences of a code" that is hard to make a meaningful difference between a weak example and a strong example?

To sum up, I will use the same selection and coding principles to compare different phases of the ongoing project process. To select the pictures to work on, among data sets, I will focus on the high definition pictures that are produced to present the human eye level, showing the outdoors especially public or semi-public spaces of the Quayside project by Sidewalk Toronto as a future Smart City. Additionally, in order to perform the content analysis more practically, I will use the software Nvivo to code, categorize my datasets and produce outputs in the forms of charts.

2. Compositional Interpretation

The compositionality of the image is one of the central methods that I will use while evaluating CGI's – actually 'what they are' and 'what they show'. I will explain in the next chapter why drawings (basically as images or visuals) are vital to the architectural representation and to the practice of architecture in general. But now I am contented by saying that my research is dependent on the visuals as direct, tangible and highly circulated documents of future imaginations, thus I take them as meaningful compositions.

This method is directly related to the site of an image itself to interpret its importance, emphasizing the compositional and also technological modality as put by Rose (2016). Rose conceptualizes this method the 'good eye', functioning like 'visual connoisseurship'. Thus, the method of compositional interpretation is investigating the following categories; content, colour, spatial organization, light and expressive content. I will explain each of the categories and use them while evaluating my samples.

One may start with decoding "the conventionalized visual symbols". Rose associates this part with the discourse analysis, in particular iconography, inquiring how social difference is constructed. This method requires familiarity with similar image sources to identify key themes, keywords or recurring items, in other words, a "broad contextual knowledge". I will depend on my professional background and education as an architect on that, but I will use the architectural representations and techniques as an interpretive way to look to the pictures content-wise – in the next chapter as mentioned.

Second, the colour that is related to realism, hence the persuasive power of an image. The choice of colour may stress some elements on the picture or may add depth to the image, for instance, by faded colours to form an 'atmospheric perspective' etc. More technically speaking, the colour of an image is the output of a combination of different aspects such as hue; the actual colours, saturation; purity of colour in the colour spectrum and value; lightness or darkness of a colour. Hereby the final combination of a particular image forms its overall harmony and effects.

The third feature to evaluate the compositionality of a picture is the spatial organization, by which an "image offers a particular viewing position to its spectator". This aspect directly relates to the techniques by which the image has been produced. Rose refers to the level of the eye as the center of the geometry on the picture; "The level of your eye is always the same as the horizon of a painting. It is also the level at which the rays of vision converge at what is called the vanishing point". Additionally, she exemplifies the use of perspective drawing as following; "using a very low eye level might represent the way a child sees the world", or "a low eye level might suggest that the painting was made to be seen from below". Spatiality is not always this simple and may include manipulations, reflections, second- or third-point perspectives instead of one-point perspective, etc.

The fourth aspect is the light that is closely related to the capacity or emphasis of the sight also the principal perception of the colour and space of the image. Finally, Rose states the expressive content, which is rather difficult to elaborate since it is linked to the 'feel' of an image. Rose cites Taylor (1957) in order to define this aspect as a "combined effect of subject matter and visual form" a more subjective or interpretational approach to analyze the content which is suggested by Rose as a crucial method but yet careful to handle with. It is also important to examine what are the technologies and/or networks are being used to produce the picture since the technological constraints or opportunities are highly effective on the finalization of the image.

3. Semiology

Semiology is presented by Rose (2016) as the most solid tool to evaluate visuals. According to her, it offers "a very full box of analytical tools for taking an image apart and tracing how it works in relation to broader systems of meaning", encompassing social and compositional modalities on the site of the image itself with Rose's terminology. Semiology, as "the study of signs", offers a "certain kind of analytical precision" depending on a specific definition of science that is challenging the ideology. Rose elaborates her points with references to Marxist theorist Louis Althusser: "Ideology is the knowledge that is constructed in such a way as to legitimate unequal social power relations; science, instead, is the knowledge that reveals those inequalities." Thus, semiology, as a critical visual method, often targets the advertisement business as the "most influential ideological forms in contemporary capitalist societies". Hall (1997) identifies the semiotic approach with "the how of representation", he continues on deciphering further, in that sense, that the poetics are "how language produces meaning", whereas the politics stand for "the effects and consequences of representation".

Even though the semiology method stands out to have a rich theoretical background, it doesn't supply an analytical approach on how to deal with visuals. Rose asserts that such study usually turns into a form of in detail study of cases studies based on relatively fewer images in numbers. However, the method advocates that there isn't any mandatory relationship between signified – a concept or object, to the signifier – a sonic or visual attribute. Thus, such a relationship is rather constructed as developed by Ferdinand de Saussure. To elaborate and illustrate this, Dyer (1982) provides a list of four categories on

how one may use human signs too in visuals with relation to representation of body (race, gender, race, hair, body, size and look), of manner (expression, eye contact and pose), of activity (touch, body movement, and positional composition) and finally of props and settings. I will use semiology as a complementary layer on my visual analysis, developing how the images are socially constructed.

Limitations

The study is objectifying several CGI's from a large-scale urban development project claiming to offer smart living environments. The working visual datasets are, in a sense, secondary data as digitally obtained images reflecting a general essence of how those images function and why they are representative or significant. As indicated by Hox & Boeijs (2005) the use of secondary data holds some issues such as obstacles of location data sources, detecting relevant and qualified data. In my case, or even before this research, I have repeatedly come across with the same visualisations in various medium related to the Quayside project. Then I searched for the sources of the images and found the official website as the source. Therefore, my research has initiated with a core idea of digital exposure to the CGI's. In this manner, the visual data sets became the secondary data for my research, still, the data is located, relevant and qualified. I strived to be reflexive on my methods and findings since the research is based, after all, on personal interpretations.

Before all, the critical visual methodologies assume that the selected visuals are important to look at in detail, this particular look may overstate certain aspects. The selection of visuals set forth a serious and in-depth investigative look that specifies "an interpretive technique" (Rose, 2016). Thus the aim is to analyse and go beyond largely circulated images and to offer different ways of seeing. For the scope of this study, the authors and developers of the Quayside project will be discussed. However, the network through which the final images have been produced remains latent. As discussed in the methodology part, the process of the production may definitely affect the outcome, yet the aim of the study is to target final images as artefacts and to focus on how they are being perceived by the public or the possible audience. Thus, the thesis is aiming to reflect beyond looking at digital visualizations but to point out a practical and critical approach for future urban thinking. Therefore the research is not limited to the visuals but more interested in how they are conceived as a case study to develop further ideas. Finally, the study is focusing on the visual analysis by purpose, however, it should be noted that the research would have highly benefited from other methods of ethnography such as observation of the current situation on the site before any construction or interviews with authors or developers of the Quayside project, as well as the people of Toronto.

Theories on Imagining the Space

How Architectural Representations Work?

Since the dawn of civilization, “drawing” has not merely meant “reproducing”. Instead, as Paul Gauguin understood (“I close my eyes to see”), it means probing our internal world.

(Belardi, 2014, s. 9)

I have drawn the detailed explication on how we may approach visuals in order to interpret them beyond being merely pictures. In this chapter, I will initiate by elaborating my approach to what type of visuals I am dealing with. For this purpose, I will start by providing the architectural theory and research that is illustrating the practical meaning of an image or a drawing – as a representation which is essential to the architecture. From this point on, I will shift my focus on the architectural image-making as persuasive documents. Those made-to-convince visuals are functioning similar to advertisements, thus, I will briefly touch upon the consumption culture and society.

Architectural Representation as Remote Control

For Baxter (2017) architecture, “like people”, is glorified as a “central element of cities”. This central element, even though it offers a considerable space to individuality and creativity, works in a well-defined conventional system which is hard to shake. Nevertheless, the conventional and practical instrument of architecture, which is the practice of drawing, is not naturally nor historically linked to the practice itself. Robbins (1994) argues that how we perform architecture today is “relatively recent and historically situated”. He continues by pointing out that “geometry, not drawing” was initially important to solve design problems or to design buildings. Robbins explains that until the Gothic period, drawings were used as documents to promote a specific architecture with aims of raising money or making a choice between building alternatives. Toker (1985), who leans his theories on a schematic elevation drawing from the 14th century³, asserts that an important split between designer and builder became apparent. As Toker articulates, the Gothic period facilitated the “design by remote control” giving the architect a “liberation from daily involvement at the construction site which fed his new and higher status”. Even so, it wasn’t a sudden withdrawal from the construction site (that is still not possible) but served as a starting point for the practice of architecture to spend more time on drawings that represent the buildings in more specific details, test their ideas through those drawings and to acquire a “gentlemanly status” (Robbins, 1994). In the Renaissance period, in line with advancements in arts, sciences and other sites through global conquests of the Westerners, architects were allowed to be more experimental and artistic. Robbins also states that with the emergence of printing, architects engaged in the “humanist quest to appropriate the classical “as a form of widespread architectural memory. By the 15th century, the popularity of the drawing even exceeded the role of scaled models to architectural practice. Thus, by the 20th century, drawings became a “natural and universal currency of architectural discourse”.

³ The Sansedoni elevation has been found in Siena, Italy dating 1340.

Previously I have argued that the visual is dominating our world globally. This means, for the field of architectural practice, a global standard for drawing is shared by almost all practitioners. Basically, the orthographic projection with plans, sections and elevations is the essential drawing instrument of an architect. Perspectives have a special position among the drawings since with the perspective the future buildings shall be represented most artistically and impressively. Robbins (1994) defines perspective drawings as follows; “drawings of solid objects on a two-dimensional surface done in such a way as to suggest their relative positions and size when viewed from a particular point”. Such drawings enable the viewer to feel as if he or she is moving around the building although this implies a kind of problematically human-centred “solipsistic relation to architectural space”, as put by Robbins. Furthermore, through the use of perspective drawings the eye became the “centre point of the perceptual world as well as of the concept of the self” (Pallasmaa, 2005), offered “a point of view into the future building” in order to “orientate and direct the attention and ‘subjectivise’ the project” (Houdart, 2008).

Architecture at the Age of Digital Image

Following the scope of this research, I will investigate how CGI’s are functioning to construct persuasive images in the realm of concept development and marketing. Later, I will focus on how those images influence the concept of public space in Smart Cities. Barratto (2017) describes CGI’s as “two-dimensional compositions usually conceived from three-dimensional digital models and often in a realistic style”. The purpose of the digitally produced images, according to him, is to “estimate the future of the work constructed within its context” not only in a realistic way but more importantly as representations of “fantastic and impossibly grand scenarios”. Architectural representations play a ‘decisive role’, as Houdart (2008) argues; “they make a whole world come alive and, at the same time, act to convince multiple audiences (in particular the clients) of its ability to function”. Rose (2016) notifies that such images were not common until recently, to “be found on the billboards of even quite small building projects”.

By all means, the digital turn has affected the practice of architecture. Graves (2012), at the conclusion of his life as an experienced architect, points this impact out as follows; “the computer is transforming every aspect of how architects work, from sketching their first impressions of an idea to creating complex construction documents for contractors”. However, he insists, perhaps nostalgically, on the role of the drawing as “not just end products” but an essential part of “the thought process of architectural design”. Riahi (2017) also problematizes the process of ‘image making’, arguing that the “shift allowed architectural projects to move from objects to systems, networks, and organisms, and therefore significantly transform the processes of thinking about and representing architecture”. The network of image-making appears as a symbolic and complex site of investigation.

Nevertheless, once more, the relationship between the architect and the built environment is dramatically being redefined through social, technical and practical terms. Tschumi (1994) declares this dialectical process as following: “Architecture only survives where it negates the form that society expects of it. Where it negates itself by transgressing the limits that history has set for it.” The visual emphasis is being facilitated by the wide use of digital software as design tools but, even more crucially, the digitalization of the practice brought along a sense of alienation from the space itself. Pallasmaa (2005) declares that this feeling is commonly apparent at, for instance, most tech-advanced settings, such as hospitals and airports due to “the dominance of the eye and the suppression of the other senses”. According to him such ‘hegemony of vision’ is depending on multiple inventions composing a network of visuals such as the printing press, artificial illumination, photography, visual poetry and the

new experience of time, resulting with “the endless multiplication and production of images”. Pallasmaa continues to associate the spread of today’s “superficial architectural imagery” with a cancerous spread avoiding “tectonic logic and a sense of materiality and empathy”. Furthermore, he argues that even urban planning has been undertaken as “highly idealized and schematized visions seen through *le regard surplombant*”⁴

Pallasmaa (2017) points out the immersion of the architectural tool: “The pencil in the architect’s hand is a bridge between the imagining mind and the image that appears on the sheet of paper. In the ecstasy of work, the draughtsman forgets both his hand and the pencil, and the image emerges as if it were an automatic projection of the imagining mind.” Such immersion is likely to be attained through digital mastery, however, typically criticized due to its impersonal attributes. On the other hand, keeping in mind new definitions of reality through Augmented Reality (AR) or Virtual Reality (VR) devices, the computer-generated images blur the border between two worlds of realities; actual and virtual. Thus, such border of representation will be hard to distinguish and locate (Mitchell W. J., 1992). Barratto (2017) describes the VR enables us to fully immerse into the un-built architecture. VR will allow “the observer to ‘enter’ into space”, this an imagined space, however, the “wandering eye is no longer limited to physical space, it is part of a whole new architecture, immaterial, intangible, but visible”. Beyond the medium at which a representation is made, how a viewer perceives the visualization of a project may be subject of confusion. Rose (2016), below, problematizes this aspect of how photography and CGI’s are intertwined:

“If such visualizations are often described as ‘photographs’ when they are reproduced in newspapers – and they often are – and if the billboards on which they appear carry warning reminding their viewers that a ‘Computer generated Image is indicative only’ – which seems to imply that they might be taken as images of real scenes – are their digital qualities irrelevant so that they should be understood as photographs after all?”

For an architect, diffusing into the instrument of representation seems to be possible now at a virtual level yet detached from familiar senses. Nevertheless, the advancement of the technology makes the estimations almost impossible on “just how computers will change perceptions and conceptions of architectural forms in the coming years” as Mario (2001) declares. No matter what technological advancements are at the stakes, architecture has a tense and critical relationship with the notion of representation. Rattenbury (2002) elaborates this tension as following: “Architecture is driven by a belief in the nature of the real and the physical: the specific qualities of one thing – it’s material, form, arrangement, substance, detail – over another. It is absolutely rooted in the idea of ‘the thing itself’. Yet it is discussed, illustrated, explained – even defined – almost entirely through its representations.” According to her, the representation, therefore, the immersion with the instrument of the architecture, will always be ‘partial’ and indirect or mediated.

Architectural Representation as Advertisement

Apart from the role of idea or concept development, architectural representations serve as the tool of communication to describe the building before it is being built. However, recently it became more common to see architectural representations in the form of pure advertisement, marketing possible

⁴ The look from above.

urban futures. Although there is a historic purpose of the drawings to convince patrons (Robbins, 1994), today with the dominant capitalist culture and neoliberal planning politics, such visuals are targeting a far larger audience – whether virtual on the internet or physical through billboards. They are made implicitly to convince the future citizens to believe in the feasibility of a producer's vision, adopt it and to act accordingly.

Urban vision today is produced at the local governance level, in accordance with the nation-state politics and the participation of multinational corporations. Harvey (1989), coins this as 'urban entrepreneurialism' having a flexible corporatist vision, performing 'inter-urban competition' where the main responsibility of governance is to ensure a 'good business climate' for the city to strive among others. He points out, in general, the crisis dependent capitalist system and more specifically de-industrialization are underlying reasons for such 'city boosterism'; "with the declining powers of the nation-state to control multinational money flows, so that investment increasingly takes the form of a negotiation between international finance capital and local powers doing the best they can to maximize the attractiveness of the local site as a lure for capitalist development." Harvey continuously relates this boosterism with a constructed 'urban imagery' that may cause a sense of alienation with over control of the urban space. According to his view, these are most possibly "ephemeral fixes to urban problems". It is a provoked consumerism in order to consolidate the idea that this place would sell for the capitalist flows. Harvey notes such consumer attractions are places for the 'urban spectacle'. Harvey points out the illustrated urban vision as something to a market where the imagined city conceptualized; "as an innovative, exciting, creative, and safe place to live or to visit, to play and consume."

The overarching frame of the 'urban entrepreneurialism' is the neoliberal planning and politics. Throughout a historical perspective, Manfredini (2018) asserts that there is still a belief in neoliberal governance to deliver "social good through a market-driven approach". However, that trust entails the risk of the abolishment of democratic planning practices. Accordingly, Baeten (2017, p. 105) argues that "The neoliberalisation of planning implies a partial retreat by planning as a public institution from its conventional core, namely the improvement of the built and natural environment through some sort of concerted effort". In this line, Cardullo and Kitchin (2018) describe neoliberal urbanism as "a model of urban growth based on marketization", remarking that the "civic paternalism and stewardship" has been re-emerged, whereas the policy-making and urban planning debates has been reduced to a state of economic argument while the citizen has been degraded to a level of consumer or user.

As a member of the consumer society, to live in the cities of capitalism means a current exposure to visuals aiming to convince us to buy something that is belonging to the future. Berger (1972) asserts that there has never been such a "concentration of images, such a density of visual messages" throughout the history whereas the practice of democracy is substituted by the promise of consumption. Berger sets forth the logic of advertisements – the process of manufacturing glamour – as following: "Publicity is about social relations, not objects. Its promise is not of pleasure, but of happiness: happiness as judged from the outside by others. The happiness of being envied is glamour."

An advertisement may be seen as a banality, a common cliché. Rose (2016) defends that the power of visual advertisement is based exactly of this humble appearance as "pretty superficial things". They may sometimes even challenge the 'reception regime of advertising', as it is apparent for instance at Benetton advertisements by asking "their viewers to engage with 'big' issues -death, violence". The founder, Luciano Benetton reports: "We did not create our advertisements in order to provoke, but to make people talk, to develop citizen consciousness." (Faye, 2014). However, it should be noted that "such campaigns do wonders for the company: a political alignment with consumers is much stronger than a strictly aesthetic one, after all" (Blickwink, 2012)



Figure 2 *The Blanket*. 1990, Photograph by Oliviero Toscani for United Colors Benetton

Advertising or branding may possibly be associated with the experience economy. Rose (2016) argues that the ‘feel’ at certain brand shops as a continuation of the marketed products engage the customers to identify themselves aligned with the so-called character of the brand – for instance, the worldwide unified architectural character of the Apple shops. She continues as follows; “the idea is that you are buying not just a product with a specific functionality but also a whole experience that you like”. The strategy of increasing the sales with the advertisement is considered to be functioning as well for selling urban development projects to future buyers marketing through “intense aesthetic of a contemporary ‘digital vision’” (Rose, 2016). Setting aside advertisements that are challenging the norms, we can notice a pattern of signs that is reoccurring through visuals. Williamson (1978) defines those signs as referent systems and identifies three major referents as nature, magic and time. Williamson explains the system of major referents as follows; “We misrepresent our relation to nature, and we avoid our real situation in time. I have placed ‘magic’ as a topic between these two because in a way it combines the other two fallacies — transforming our temporal relationship with nature.”

I would argue that those referents are also explicitly present in CGI’s for the purpose of convincing the viewer into the designed future. Architectural representations use high-definition and photo-realistic techniques for that conviction. Nevertheless, collage techniques have also played a significant role in the image-making. Such electronic collage “allows ready combinations of synthesized images with captured ones – to place synthesized images in real scenes, or real objects in synthesized scenes” (Mitchell W. J., 1992). This is a method of ‘digital cut-and-paste’ results as persuasive images through skillful digital manipulation that merges the real with the computed. The images are hybrid of multi-layered elements such as photography, CGI’s, visuals taken from online databases. Thus, as Houdart (2008) argues that they; “are not photography per se but seem to borrow some of its qualities and characteristics”. However, Iliescu (2008) protests such digital montages as they are “antithetical to both the original intent of modernist collage and its subsequent role in postmodern critique”. According to her, that original intent of the modern was to challenge the border between art and life by surprising the viewer. Such kind of collages might, “evoke thoughts and emotions outside the strictly aesthetic realm” through ‘unexpected associations’.

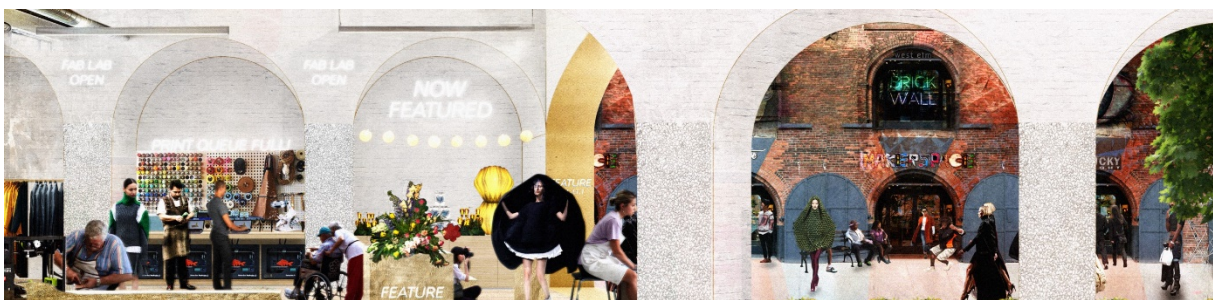


Figure 3 *A clookage, the Street Life after Retail* by Matthew Ransom and Violet Whitney for Sidewalk Labs

Three Accounts from Architectural Ethnographies and Conclusion

The theoretical frame that I have so far referred is covering the role and significance of the architectural representations, in order to reveal the current vision for Smart Cities. I have argued that architectural representations in the form of CGI's became a sort of commodity. Pallasmaa (2005) describes the current situation as a “crisis of representation” in architecture implying an affinity to larger attributes to the postmodern term – a struggle over the naturalness and innocence of ‘reading’ (Ebert, 1986). The practice of architecture is certainly pertinent to the act and purpose of image-making – laying deep down in the very essence of the profession. As shown below, the aimed visual quality and targeted audiences are not always aligned with the final performance of the building in a real city. In some cases (Figure 4-5) architectural representation outshines the actual building, “surpasses the architecture itself” (Rattenbury, 2002). To a degree, it is hard to attain the ‘feel’ of the presented CGI, thus overruns in construction budgets and schedules are almost unavoidable. On the other hand, on rare occasions (Figure 6-7), the atmospheric effect aimed by the CGI's is achieved or even outperformed. It may be, definitely, argued that the obtained ‘feel’ of the building is also ephemerally bound to the event or quality of the actual building maintenance. What is more, the software used in representations play a crucial role at the quality of final outputs, even so, the point is the critical alignment of the pre-image of a building that is digitally produced with the final product as the building itself.

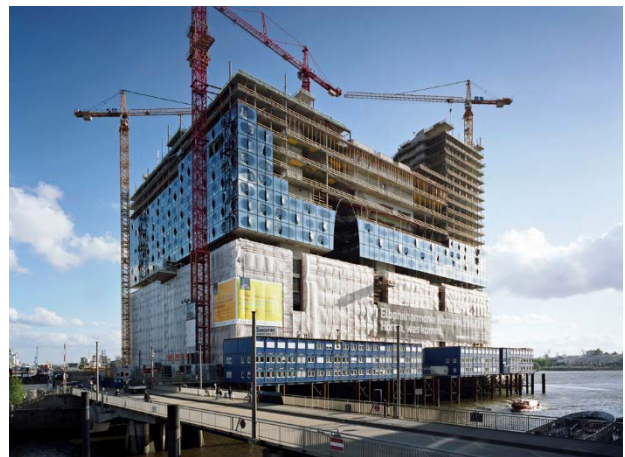


Figure 4-5 CGI from Elbe Philharmonic Hall in Hamburg by Herzog & DeMeuron Partnership and photo during the construction by Oliver Heissner

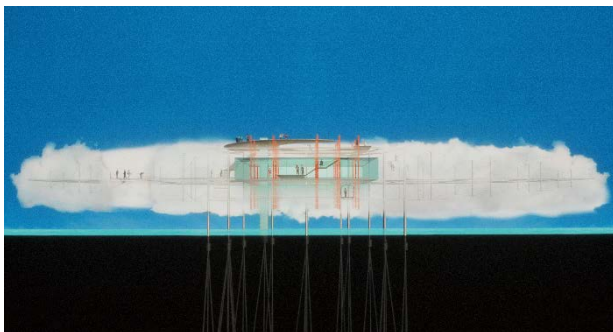


Figure 6-7 Drawing for the Blur Building by Diller Scofidio + Renfro and photo by Beat Widmer and Dirk Hebel

With a particular focus of my research, I chose to look at architectural representations of Smart Cities as ethnographic documents, witnessing the imagined future. This approach stands out to be unique in a sense, nevertheless, I found similar approaches to evaluate and understand CGI's as sites of meaning production. Those accounts investigate the alignment of the digital pre-image of a building with the final product as the building itself. Following, I will conclude three accounts through rather long quotations in order to illustrate the ethnographies closely, followed by my own interpretations.

1. Made by the Office for Metropolitan Architecture: An Ethnography of Design

The entire book is dedicated to an ethnographical project, where Yaneva (2009) is 'following architects at work'. This book provides an intimate and in-depth yet objective analysis of how one of the major architectural offices (OMA - Rem Koolhaas) is functioning. The following is from the introduction of the book.

"This book presents an ethnographic account of the design rhythm in the Office for Metropolitan Architecture. Written as a collection of short stories, it draws on the mundane trajectories of models and architects at OMA and shows how innovation permeates design practice and everyday techniques and workaday choices set new standards for buildings and urban phenomena. In these stories of the invention the 'Eureka!' moments are missing. They are replaced by routine gestures of model making, recycling, assembling, recollecting, rescaling. This inquiry into architecture-in-the-making is based on participant observation in OMA, extensive interviews with architects, and photo documentation on various projects: the Seattle Public Library, the Whitney Museum of American Art in New York, the China Central Television (CCTV) in Beijing, the Los Angeles County Museum of Art (LACMA), the Casa da Musica in Porto, and others."

Frankly, the visuals are not the only ethnographical documents to deal with. However the author, as an outsider, is after revealing the way of how buildings are 'obtained' instead of being 'projected' or 'anticipated'. Yaneva (2009) highlights this as the "way a building first becomes visible, present and real in the office". She continues "through the trials with visuals the building becomes more and more thinkable, more 'obtainable'; it is possible to get it, to achieve it." Accordingly, the incrementality, experimentality and play of the accident hold a large space into the design process; "a building is not obtained in an astute double-click moment of invention, but through numerous little operations of visualization, scaling, adjustment of materials and instruments." For Yaneva what is characteristic to OMA is the difference between 'architecture in the making' versus 'architecture made'. This attitude is parallel, at least at a theoretical level, to Koolhaas' ideas on rationalized modern urban environments that are sharing the "blessing of science, universally". Koolhaas subsequently asks: "What remains of the city once, the unpredictable has been removed" (Koolhaas, 2002).

Yaneva reflects upon how design functions, defending no foundational agencies for the act of design but "something counteractive". Thus she further argues, in relation to the practice of OMA, that the design is to re-design; "design relies on a cognitive and experimental move of going back, rethinking carefully and recollecting, re-inventing, re-interpreting, re-looking, re-doing everything once again in a new combination of conservation and innovation". Thus, the way that is preferred in order to approach architectural representations does not appear to be in strictly defined forms of drawings or visuals, but through a manifold of mediums to obtain a building or an urban environment.

2. Networks, interfaces, and computer-generated images: Learning from digital visualizations of urban redevelopment projects

In this research Rose et al. (2014) are investigating how the 'software-supported space' is produced, by doing so they are approaching to CGI's not just as images but as 'interfaces' focusing on the circulation through 'networks' of image production or image-making. The authors are explaining how they refer to 'interfaces' – the intersection between two objects or systems – as following: "In digital studies, use of the term is often restricted to the relation between a human and a digital device, or, to be more specific, to contact between eyes, hands, screens, mice (or graphics tablets), keyboards (or gaming control pads), and the software driving the hardware." Thus, they use ANT principles to show "less on what they show" but focusing on "how they are made to show". Their research is based also on ethnographic research to slice down the network that is projecting CGI's for Msheireb Downtown development project in Doha by a group of international consultants; AECOM and Arup with architects Allies and Morrison from London. The following is from the research, illustrating the 'ecology' of the CGIs:

"There is a further range of site wide and executive consultants with specific responsibilities, coordinated by the Master Development Consultants (MDC) team, and including Executive Consultants (EC), Executive Architects (EA), and landscape architects among others. Nine design architects (DAs) were chosen to design the hundred or so buildings on the site. A partner at Allies and Morrison was appointed as the Architectural Language Advisor (ALA) to the project, and has been a particularly enthusiastic advocate of the project's CGIs as a means of cohering the project's different components into, as he said, "something one wants on the cover of a magazine". Here, then, we have a developer of a large-scale urban redevelopment project investing significant time and money in creating a suite of CGIs that will picture "how that place will look and feel" (MDC manager) when it is finished."

Rose et al. capture the temporary nature of the visuals, whereas the final set of selected images are definitely 'seductive' mobile and easily alterable. The players of the network of image-producing are acutely aware of the 'audience', authors report that "people tend not to look at the architecture but at the image and make decisions based on liking or not liking the image". There is a subtle alienation of the architectural practice compared to processes of 'obtaining a building' at OMA, as mentioned previously. Through the strict definitions of the CGI network, every player performs what it is expected; architects of the development projects are after the 'rightness' of the building. On the other hand, visualizers, who are professionally producing the visuals of the project, aim a 'striking' look, working perhaps with a 'render farm'⁵ from China, whereas ALA is seeking out for "magic moments". I have mentioned the magic previously as one of the major referents in the advertisement industry. Williamson (1978) argues marketing requires a magical attribute by definition when it is deemed as a miracle "we do not feel we need ask for an explanation". In contrast, particular to this case but also emphasizing the alienated nature of the network, the client insists that the CGI's have to "look distinctively Qatari" to be more persuasive for the final audience of the project, in contrast with the fact that the image-makers doesn't have much familiarity with the context of the project (Rose, Degen, & Melhuish, 2014).

⁵ Render farms are mainly situated in developing world, mainly in China, providing fast and cheap render services through digital delivery thanks to massive servers but also cheap electricity as Rose points out.

3. Copying, Cutting and Pasting Social Spheres: Computer Designers' Participation in Architectural Projects

The final account is based on the ethnographic fieldwork at Kengo Kuma's office, focusing on how the visuals are produced for the Japanese World Fair in 2005. The task is to protect the site of the project and at the same time to design something that is 'visible'. Kuma's practice chose to approach the problem via a more abstract manner with visual representations, closer to modernist collage instead of photo-realism. Houdart (2008) reports from image-makers that are working with Kuma's office that, even though it is possible to make the drawings as "super-realistic looking images", the visuals especially showing people often lose the level of reality due to the "over-standardization" of human elements – somehow "resistant to drawing". Houdart, in contrast, uses the term 'cosmology' that is the natural order of the universe for the creation of visuals.

"While looking at the actual operations which constitute this representation technique, I would like to show how these virtual montages feature as cosmologies in the making. Here, 'world' or 'cosmology' are not simply manners of speaking. They must be seen as common spaces to design, for which the cohabitation of such diverse things as human beings, buildings, roads, trees, skies, cars, and their respective ways of existing must be anticipated. This is a serious matter: Architects, while designing, digitalizing, copying, and cutting and pasting images, manipulate social spheres and give birth to new ones by testing and submitting new social configurations."

The final output, as sets of visuals, is more than images but a projected cosmology or parallel cosmologies that directly aligned with myths, ambition and character of a group of people, in this case targeting Japan as a nation. Houdart witnesses through reoccurring images (same young couples with children or babies) that it is part of a vision that is crediting "an already well-established social order based on a reiterated family contract". This cosmological approach is clearly linked to the task of the project of Expo, nevertheless, I would argue that such ideological purpose of the visual is essential for any CGI's to represent or to create an identity for the audience. This is 'ideology as overlay' as introduced by Williamson (1978), conceptualizing the ideology as the overall conditions of the society in an imaginary form. A hard-to-keep promise for a space to belong or to identify with. Williamson concludes as follows; "All of us have a genuine need for a social being, a common culture. The mass media provide this to some extent and can (potentially) fulfil a positive function in our lives." In the case of image-making through CGI's, the cultural identity and social structure of the close future are being implicitly yet constantly mediated to masses.

Critically, Houdart (2008) argues that no matter how it is composed what makes the conviction stronger in architectural representation is not the precision or tenacious care in details but the expressive or artistic element "to transport the client away from his world into a new one; it is subversive almost by definition". Thus the drawing doesn't have to "stick to reality" but to "do something" in order to "show that the newly designed world would 'work'".

Digital Transformation of the Public Space

"The Goliath of Totalitarianism will be brought down by the David of the microchip."

Ronald Reagan, June 14, 1989

Having a close look at the architectural representations as persuasive visuals targeting a future society, I will orient my focus into what is the physical space that is represented and imagined by those CGI's. Through the specified aim of the study, I will briefly summarize the notion of public space in Smart Cities that are to come forward. Thus, this chapter will be about how the built environment, public space in particular, has been evolved following theories and ideas on Smart Cities and their transformative potential. In this manner, the central focus of this theoretical part will be about design practices and imagination in mutual relationship to the production of public space.

Defining the Public Space

Urban public space is the core of the urban activity, holding a notion that has political, socio-economic and historical connotations and meanings. Sennett (2017) basically defines the public space as "a place where strangers meet", emphasizing both its anonymity and its emancipatory character. Dovey (2016) conceptualizes public spaces as the "primary site where a sense of the 'common' becomes embodied in everyday life before it becomes 'community'". Above all, the term public space implies a space reserved for the public itself. In line with this, Iveson (2007) asserts that public spaces are intuitively representing "the triumph of the public over the market". According to him, such spaces are 'ceremonial spaces' that are enabling both celebration and protestation. However, the main motive behind being present at the public spaces is to be "included in the public represented through the space". Therefore the uses and the design of the public sphere are essentially political as well as subconscious. (Benjamin, 1935-1999). The aggregating power of public space that was assumed to be powerful is dissolving. Sorkin (1999) summarizes the modern city planning as a 'conflict avoidance' activity where enemies of public space are privatization, identity politics, and sprawl. Through the 'anti-urban character' large-scale suburbanization, gated communities, or high rise buildings the interaction of the street level is diminishing and it is leaving its place to a more segregated and polarized urban life (Dovey, 2008).

Lefebvre (2005) approaches everyday life critically to find out a hint for 'knowledge of society' in relation to its spaces. Lefebvre argues that daily life is the product of the mode of production, includes repetition, relates with the context of time and space, does not overlap with lived experience, encounters use and exchange, connected with 'needs-desires-pleasures', linked with the body and contains the extraordinary. In that sense, everyday life is the "most familiar" yet the "least known" covering a network from the banal to the festive. Accordingly, King (1980) associates the modes of production with the orientation of man, he relates pre-industrial era with 'nature's time' which is task-oriented, and instead, he aligns the industrial revolution with clock-oriented time. Thinking the transformation of the public spaces through the artificial revolution or the fourth industrial revolution, Schwab (2019) draws attention to the new dimensions of the proliferation of digital goods and services. As much as the dominant modes of production and politics formed the everyday life throughout the course of history, the fourth industrial revolution would, or even already begin to, challenge the ways we collaborate, produce, consume, communicate and make sense of the world.

Smartification of the Public Space

According to Townsend (2013, p. 15), Smart Cities are places "where information technology is combined with infrastructure, architecture, everyday objects and even our bodies to address social, economic and environmental problems". Karvonen et al. (2019, p. 1) define the Smart City as simply as "improving and optimizing urban services through the application of ICT". Today, we are, yet partly, living in Smart Cities (Karvonen, Cugurullo, & Caprotti, 2019). The tendency to rely on data, in order to render the urban environment more rational, predictable and efficient, most directly aligns with the logic of modernization. Digital applications first appeared to be bound to the virtual, to the cyberspace. The public space of interaction and exchange found a new, free, anonymous territory of expansion – an 'electronic agora' that "subverts, displace and radically redefines our notions of gathering place, community and urban life" (Mitchell W. J., 1995). The sudden and extensive propagation of computers altered the nature of interaction with our environment. Koolhaas (2002) identifies monitors and screens with alternatives for conventional windows; "real life is inside, while cyberspace has become the great outdoors". Nowadays the assets of the digitalization transcend the boundaries of the virtual space. Following the extensive developments in digital technologies, smart applications are emerging to optimize urban services and environments. Townsend (2013) names the smartphone as a "megacity survival kit, a digital Swiss Army" or Greenfield (2017) delineates them as "universal, all-but-indispensable mediator of everyday life". The disruptive use of smart mobile phones supports the common views on the privileged and alienated visual world through 'technological miniaturization'. Anceschi (1996) pinpoints the emergence of objects that are 'black-boxes' "richer in functions but poorer in transparency". This pattern is arguably featured for the design and conception of Smart Cities.

Smart City technologies that are being used today are infused, not necessarily visible, to the existing urban infrastructure and services, and are therefore hard to concretize. Even so, day-to-day urban life is getting more dependent on the conveniences of smart applications through layers of digital infrastructure. By virtue of the recent change of paradigm, the digital began to determine public life on a large scale. Amin and Thrift (2002) argue that we are provided with the unique opportunity to be actively present at a set of activities at once without changing the location. The city, "a vast narrative structure that constantly re-presents itself", is becoming "something that is beginning to read us" through constant computing. (Amin & Thrift, 2002). The places in Smart Cities of that sense are 'code/spaces' where "software and the spatiality of everyday life become mutually constituted" (2011). On the other hand, Graham (2005) asserts that 'coded worlds' or 'computerized spaces' operates beyond the virtual, exerting "their power over the geographies and lifeworlds of capitalism". According to Graham, aiming for a more secure, computed society that is theoretically freed from risk, crime or congestion, entails the risk to avoid contact with the 'failed consumers'. Whereas, to meet the 'other' is basically the essence of public space.

'Smart' at the Public Space

Owing to the emergent municipal boosterism and urban competitiveness, Smart City became a label that numerous urban actors are willing to embrace (Karvonen, Cugurullo, & Caprotti, 2019), however, there is no such place that we may observe a Smart City as a whole yet (Townsend, 2013). Nevertheless, academic circles indicate two distinct built examples embracing Smart City discourse; Songdo in South Korea and Masdar in UAE (Sennett, 2012); (Hollands, 2015); (Albino, Berardi, & Dangelico, 2015); (Kitchin, 2015). Sennett describes Masdar as follows: "There's no stimulation through trial and error;

people learn their city passively. 'User-friendly' in Masdar means choosing menu options rather than creating the menu." Whereas Calvillo et al. (2015) describe Songdo as follows; "a system that theoretically continues to produce wealth-without-end through the construction of huge conduits for bandwidth and of vast quantities of environmental sensors, all focused on the monitoring and indexing of its inhabitants' online and offline behaviors." All in all, it is clear that the Smart City frame has to be understood as the continuation of neoliberal discourse or neoliberal planning and urbanism. As Harvey (1989) argues, it is hard to deny that what is rising is "the postmodern penchant for the design of urban fragments rather than comprehensive urban planning".

There is an underlying interplay between Smart City and the public space. The medium and experience of the public space have been expanded by means of the digitalization. There are numerous smart applications that are shaping the urban space, targetting an optimized urban life thorough vast application of ICT's, in most of the case through mobile smartphones. I determined four types of smart applications that are increasingly in use defined by whom or what it is introduced; state, municipality, design, and agency. First, the state-introduced smart where national or supranational governments are using online education tools to educate probably technologically illiterate masses. For instance, the Finnish Government issued a comprehensive free online course titled "Elements of AI" where the goal put as to "demystify AI"⁶. Second, the municipality-introduced smart where the local governments enable their citizen to engage more with their cities and stay connected with a wider network of cities. There are many examples of such municipality related initiatives⁷. The third is the design-introduced smart where architectural design practices lead innovative ways to enhance citizen engagement through smart applications. For instance, early this year White Arkitekter held a student innovation challenge accordingly and chose the application "Engage 4 Change"⁸: "The concept is presented as an easy to understand app where people can learn about planned development projects and get engaged in various ways. The underlying platform shows great promise for actors in city development to interact with the citizens they affect and make more informed decisions." Finally, the agency-introduced smart where smart applications are used exclusively to keep urban space safer for all. That type of application is passing beyond being only profitable, but they use the digital tools for empowerment by supporting the inclusivity of the city. To illustrate, the SafetiPin⁹ from India is a widely and effectively used security-mapping application " that aims to help women stay safe by letting users rate streets and areas for safety criteria such as lighting, visibility, people density, gender diversity, security and transportation. It also aggregates safety data, partly provided by its users, for use by local government and planners"

Even though the concrete physical space is being transformed and augmented by digital applications, there is also a growing awareness coupled with the counteraction against threats of digitalization that is repositioning the purpose of technology. Following the same manner, for instance, Copenhagen Catalogue¹⁰ manifests as following: "It is time to take responsibility for the world we are creating. Time to put humans before business. Time to replace the empty rhetoric of 'building a better world' with a commitment to real action. It is time to organize, and to hold each other accountable." Mattern (2016), at his point, asks anxiously: "How will the new methods of measurement and planning inform what it means to be a citizen in a quantitative community?". On the point of citizenship Işın (2017) describes it as following; "a legal institution governs who may and may not act as a subject of rights, within any given polity. Being 'a subject of rights' means having both the capacity and authority to exercise rights and duties". Thus smart or digital citizenship is a performative institution that requires action rather than

⁶ <https://www.elementsofai.com/>

⁷ For instance look at <http://nyc.pubcollab.org/>

⁸ <https://whitearkitekter.com/se/nyheter/appen-engage-for-change-vinnare-i-white-innovation-challenge/>

⁹ <http://safetipin.com/>

¹⁰ <https://copenhagenletter.org/>

just “following the script as smart citizens”. Thus, accordingly, for smart citizens, cyberspace is a “relational space in which digital citizens come into being through digital acts” (Işın & Ruppert, 2015). That cyberspace is not separated from the real nor is it an independent space of freedom. The cyberspace is governed by the code. But there is a dilemma, which is potentially relevant to the ‘code/space’ of smart cities as asserted by Işın and Ruppert; “one obeys the laws as code not because one should; one obeys...because one can do nothing else”. Thus, afore-mentioned quantitative communities become extremely vulnerable to algorithmic biases or discrimination by design.

The field of digital technologies is immense, exponential and promising, that opens up new horizons, literally augmented spaces, to perceive the new definitions of citizenship, participation and democracy in conceivably creative ways. Likewise, such an approach spatially holds, “a real possibility to approach a dynamic re-composition of spaces, places and territories articulated and influenced by information technologies at many different scales.” (Duarte & Firmino, 2009). The direction to which the future Smart Cities and their public spaces will evolve is ambiguous, nevertheless, it is evident that it will change and be re-defined. Greenfield (2017) emphasizes the urgent need to develop an understanding of how those technologies work and to ask rightful, critical questions whenever we come across with new uses of such technology. Greenfield’s call for a new understanding is crucially and fundamentally applicable to the practice of architecture and urban planning since the cities of the future are most contingent areas of technological interventions.

How to Evaluate Smart Public Spaces

I have illustrated briefly the critical role of public space in Smart Cities by explaining what I refer as public space, then I explained how Smart City concept functions with smart applications and finally pointed out how Smart Cities may possibly host public spaces in their physical urban environments. Here I will draw a concluding theoretical frame on how might we conceive of Smart Cities. With regards to this framework, I will set two sets of rules that will serve as coding categories to evaluate the CGI’s from the selected Smart City case in the upcoming chapter. The set of rules are not directly allocating how a Smart City should be, rather they are pointing out in a good faith how a livable, human-centred ecological, in short, sustainable city should be. Coding categories are providing the base to answer my essential research question: How the public space is represented in Smart City projects?

Imagining Smart Cities: Prescriptive versus Coordinating

Richard Sennett is a sociologist, planner and an urban thinker mostly involved academically with the themes of public and private identity and practice in the cities. Latham (2017) asserts that Sennett’s main contribution to Urban Studies is the idea “that the design and architecture of cities, along with the rituals of their use play a central role in the formation of society’s social and political culture”. In his book *Building and dwelling: Ethics for the City* Sennett established a unique and critical stance towards the Smart City discourse, which will also serve as the overarching frame of this study.

Sennett addresses the Smart City concept through a frame of duality. This frame mainly favors the ‘open city’ against the ‘closed city’ due to the “over-determined form”. Essentially this dual frame is part of an understanding that he has been developed over the years. In his essay, *The Public Realm*, dating back 2009, Sennett (2017) argues basically that “the closed system has paralyzed urbanism, while the open

system might free it". For Sennett 'open' is not a measure of aesthetics but rather a quality of the urban form. Departing from that point, for the case of Smart Cities, Sennett (2018) has associated the closed form with 'prescriptive Smart Cities' that "does mental harm; it dumbs down its citizens", in contrast, to open form associated with 'coordinating Smart Cities' that "stimulates people mentally by engaging them in complex problems and human difference". The main point that Sennett insistently opposes is the overdetermined urban form or the 'tight-fit'. He conceptualizes the aim of the "sheer efficiency" as an unbalanced attribute of Smart City endeavors, where the "the prescriptive city becomes unbalanced in divorcing functioning from questioning". Instead, the technology has to be used in order to coordinate rather than control, where the produced data is limited and un-purposed while comprehensive participation and decision-making are truly enabled. In coordinating Smart City model the digital processes are transparent yet people are encouraged to get involved with the data processes by interpreting and acting on it. Thus, this is a 'performative citizenship' model.

In his book *Building and dwelling: Ethics for the City* five open forms are listed pointing out coordinating Smart City. First, 'synchronic space' where several activities are happening at the same time as it is at an agora or a bazaar. This is an "invitation to mix rather than impose mixing", offering a "spatial experience both stimulating and disorientating" – that requires orientation. Second is the 'punctuated space', where Sennett uses allegorically the punctuations marks to conceive the urban space as a meaningful language with orientation points. Third, the 'porous space' comparing the built environment with organic forms that have a metabolism. He continues to define porosity as follows; "there is an open flow between the inside and outside, yet the structure retains the shape of its functions and form". Forth Sennett sets forth the 'incomplete space'. An incomplete form is not a 'shell' even though shells would provide unexhausted possibilities and porosity but rather a 'type form'; "a piece of urban DNA which takes on different shapes in different circumstances". Within its own constraints, variations and improvisations are possible – a loose-fit. Finally the 'seed planning' where the shifting complexities are coordinated. Sennett criticized master planning approach that "divides a city up into a closed system where each place function relates logically to other places". In contrast, he suggests 'farmed cities' which is conceptually functioning as "an initially unrealized, incomplete form – a seed" that is provided with "the time to grow into its surroundings." Thus seed-planning is dynamic and does relate the form of an urban project with the function at a minimum level.

Coding Categories

1. Richard Sennett's Incomplete Form

The five open forms that I have mentioned above are not identical with the "incomplete form" that Sennett's introduced at his article titled *The Public Realm*. Nevertheless, I will use five categories of the 'open form' from that article and formulate coding categories accordingly. Richard Sennett (2017) suggests to aim for an incomplete form, not a formlessness yet an openness in the structure of cities: "Buildings left incomplete, partially unprogrammed, are structures which can truly be sustainable in time; the flexible building would help end the current wasteful cycle which marries construction and demolition." Such urban imaginary comprises of urban environments that have several criteria. First, he favors the resilience and adaptability of the places and buildings rather than being quickly perishable and replaced by demolition. Second, he emphasizes the physical porosity and resistance for being 'truly urban', underlying the modern urban instrument that constructs boundaries rather than borders. Then, Sennett points out the 'spectator architecture' that incapacitated public interaction while multiplying exclusion and also possible evictions. He proposes a closer look at the neglected 'edge condition' while he argues that there is too much significance put on the design of the center. Finally, Sennett points out

the diversity or the cosmopolitanism at the street level. These criteria enable an ideal public realm by enabling people to interact with each other. This ideal praises “stimulation by the presence of others” as well as “sense of comfort and security in the midst of strangers”. However, such an ideal public life does not necessitate the identification with the crowds, instead offers the anonymity. His criteria for the incomplete form is listed below in order to frame the first set of coding categories. I have arranged sub-categories to be more specific while looking at the visuals.

- Resilience
 - Adaptive building features
 - Environmental vitality
- Porosity
 - Ground floor permeability
 - Facade material
 - Pedestrian use
- Spectator architecture
 - Interaction
 - Passive activities
- Cosmopolitanism
 - Different age groups
 - Different ethnic backgrounds
 - Different sexual orientation

2. Jan Gehl’s Twelve Quality Criteria

Jan Gehl is an urban designer from Copenhagen who has developed a unique human-centric approach to urban life and contributed to better functioning public spaces worldwide. As well as the projects and lectures, Gehl published his findings throughout his career elaborating and sharing his planning principles emphasizing the “human dimension”. According to Matan (2017), Gehl’s urban design theories are “maximizing the diversity of exchanges possible while bringing city users’ everyday needs to the forefront of planning and design concerns”, as a reaction towards the mainstream modernist emphasis on design and function. However, Matan concludes, besides the solid contribution, Gehl has been criticized due to his somehow ‘cosmetic’ approach disregarding social inequalities. In his book, *Cities for People* Gehl (2010) unfolds ‘Twelve Quality Criteria’ with three main themes (protection, comfort and delight) for “best functioning city spaces” as following:

- Protection
 - Protection against traffic and accidents— Feeling safe
 - Protection against crime and violence — Feeling secure
 - Protection against unpleasant sensory experiences
- Comfort
 - Opportunities to walk
 - Opportunities to stand/stay
 - Opportunities to sit
 - Opportunities to see
 - Opportunities to talk and listen
 - Opportunities for play and exercise
- Delight
 - Scale
 - Opportunities to enjoy the positive aspects of climate
 - Positive sensory experiences

Some of the principles listed above are certainly overlapping, but still, they form a solid work frame with essential points. Gehl particularly insists that all of the criteria have to be met in order to enhance public space: “Nothing must be left out.” Thus I will use and test Gehl’s practical ‘Twelve Quality Criteria’ to complement broader and abstract categories of ‘Incomplete Form’ from Sennett¹¹.

Quayside by Sidewalk Toronto

Presentation of the Case

Sidewalk Labs has been founded in 2015 as a sister company of Google, belonging to Alphabet Inc. family. Google essentially and initially functions as a company to provide for and enable the online platform economy; “online structures that enable a wide range of human activities”, structures that “provide an infrastructure on which other platforms are built” (Kenney & Zysman, 2016). In other words, Google’s core business is related to provide digital platforms potentially enable their users to build even more. Van Dijck et. al. (2018) emphasize the economic advantages as following: “Individuals can greatly benefit from this transformation because it empowers them to set up businesses, trade goods and exchange information online while circumventing corporate or state intermediaries”. Whereas Alphabet-Google, Apple, Facebook, Amazon and Microsoft are listed big tech companies ruling the ‘Western platform ecosystem’, promising “personalized services and contribute to innovation and economic growth, while efficiently bypassing incumbent organizations, cumbersome regulations and unnecessary expenses” (Van Dijck, Poell, & De Waal, 2018).

In March 2017 Sidewalk Labs has responded and won the international “Request of Proposals” issued by Waterfront Toronto which was established by the Government of Canada, Province of Ontario, and the City of Toronto in 2011 (Sidewalk Toronto (a), 2018). At the official web site the project is summarized as follows:

Sidewalk Labs is designing a district in Toronto’s Eastern Waterfront to tackle the challenges of urban growth, working in partnership with the tri-government agency Waterfront Toronto and the local community. This joint effort, called Sidewalk Toronto, aims to make Toronto the global hub for urban innovation.

The visual investigation of this research will be based on the current and controversial Quayside urban development project by Sidewalk Toronto. I assert that the Quayside project stands out to be the pioneer example of Smart City vision since it is developed by one of the biggest tech companies of the platform ecosystem. Through the agency of Alphabet Inc., smart applications offered by online platforms are aspiring to infiltrate into the built environment. This represents the core idea of Sidewalk Labs’ vision, in other words: “It is no exaggeration to say that software was formerly embedded in things, but now things—services as well as physical objects—are woven into software-based network fabrics.” (Kenney & Zysman, 2016)

At the official website, it is mentioned that the Quayside is the beginning of Sidewalk Toronto stands out to be "North America's largest areas of underdeveloped urban land" covering a plot of 325 hectares. The winning proposal persuasively features an adaptable pedestrian-centric layout with mixed uses through sustainability, affordability and smart mobility. But most substantially it sets forth the digital

¹¹ The full expansion of each category may be seen at the Appendices

layer that; “allows efficient operation of the neighbourhood by providing a single unified source of information about what is going on” (Sidewalk Labs, 2017). The Quayside phase would serve as a test bed to try out urban innovation to scale up in the upcoming phases of the project.

Daniel Doctoroff, the CEO of Sidewalk Labs, complains about the slow growth in urban innovation “despite the rise of computers and the internet”, also he highlights “technologist-urbanist divide” which has a retarding effect on expected development. He admits that they inspired from “great platforms like the web” and continues; “we imagined a flexible physical layer (such as street grids, open utility channels, and upgradeable digital infrastructure) with adaptable software (such as privacy rules, regulations that lay out approaches to city management, and principles of governance) that would empower people to build and change “applications” much faster than is possible in cities today.” (Doctoroff D. , 2016). Nevertheless, Doctoroff is attempting to assure that Sidewalk Labs will incorporate upgradeable buildings, customization, transparency, local community feeling, unpredictability, experimental character, local economic growth, jobs opportunities, schools, civic participation and more. In his own words, Sidewalk Labs has been imagined or designed to be an urban laboratory “showing the world what’s possible”. He continues as follows to describe it: “A large-scale district holds great potential to serve as a living laboratory for urban technology — a place to explore coordinated solutions, showcase innovations, and establish models for others to follow.” (Doctoroff D. , 2016).

Starting from the public announcement, Sidewalk Labs has prioritized the civic engagement of the project throughout 2017 and 2018 with public talks, roundtables, neighbourhood meetings, workshops, design jams, civic on-site lab discussions, pop-up stations, reference panels scattered to the city, finally fellowships and small grants programs. Furthermore, all of the activities and researches have been documented and posted online in order to keep the engagement and curiosity towards the project even larger and achieve previous promises.¹² Woyke (2018) interprets the Alphabet’s intentions as follows; “it can buck the trend by working closely with the community and tailoring the technology to local needs”. At the verge of such urban digital migration, Kenney and Zysman (2016) point out that the possible opponents are people who believe to become victims of the change unless, “if they believe they will be beneficiaries, they may help facilitate the shift.” In other words, for a good or disguised intention, civic engagement is crucial. The city conceived and promoted by Sidewalk Labs as an entity that is flexible and customizable. A commodity similar to smartphones that are being used disruptively different compared to its predecessor mobile phones. Woyke points out that “Sidewalk thinks of smart cities as being rather like smartphones. It sees itself as a platform provider responsible for offering basic tools...much as Google does with its smartphone operating system, Android.” (Woyke, 2018). Correspondingly Rohit Aggarwala, head of urban systems at Sidewalk Labs, (2018) declares Sidewalk Labs’ big ideas about cities as follows; “a highly flexible built environment that can be changed very quickly to accommodate new uses... it should allow the city to constantly keep relevant.”

The ongoing Sidewalk Toronto project, thus, is compromising of a masterplan that accommodates numerous sub-projects being tested holistically with the involvement of the private local companies, the citizen board, and academia. The related projects have been listed and shared online as podcasts that explore ideas “that will transform our cities” so far as following; mass timber as the structural building material, modular pavements that are able to be digitally controlled for lighting, heat and likewise, adaptive traffic lights that are prioritizing the pedestrian, robot delivery as automated logistics approach exercised night-time and finally promoting a new approach to waste management through clean recycling using computer vision and behavioural science feedback loop to change waste and recycling habits.¹³ Sidewalk Labs co-worked with many independent businesses. For instance, for the

¹² <https://sidewalktoronto.ca/get-involved/>

¹³ <https://soundcloud.com/city-of-the-future>

emblematic feature of sidewalks, they have collaborated with Carlo Ratti, a Torino based architect as well as a researcher at MIT's Senseable City Lab¹⁴, under the name of Dynamic Street. These modular hexagonal shaped pavement units are easily replaceable, heatable and offers dynamic use with controllable led lights. This sub-project was claimed to be tested out on site in summer 2018 to get feedback. (Rieland, 2018). Recently Sidewalk Labs has launched a basic application titled Common Space based on the open data standard Public Life Data Protocol mainly developed by Gehl Architects to enhance and ease 'public life studies' to "collect data in the field and later organise and share the data on a public database". (Risom, 2019). Such ethnographical approach understood by Sidewalk Labs as following: "Without public life studies, managers are left to steer the design, programming, and maintenance of spaces without full knowledge of what is happening on the ground or what communities need or want." (Pandey, 2019) The cooperation seems trouble-free today, however it is hard not to remember Jan Gehl's previous approach to the notion of smart, where he has answered during an interview as following: "Whenever you hear the word 'smart', beware, because that is somebody who wants to sell as many millions as possible of some new gimmick. And he is not necessarily giving you a better quality of life." (Gehl, 2016)

The Criticism

Google has a vast array of operations with an extremely beneficial business model. Considering Google's current status and explicit future ambitions, it is not a surprise that Sidewalk Labs, as an urban development initiative, has been regarded with deep suspicion by many. Sidewalk Labs first started their practice by establishing the Intersection¹⁵, a technology company to provide fast Wi-Fi infrastructure for New York with LinkNYC. Secondly, Sidewalk Labs consulted ten different cities at the Federal Department of Transportation's Smart City Challenge¹⁶ where they were working on "creating a platform that will enable cities to use real-time data to understand street activity" (Doctoroff D. , 2016). Collecting new ideas from on-going challenges, Sidewalk Labs was getting ready for a more comprehensive project, although the criticism and suspicion were not already absent. Pinto (2016) criticise Google as flowing: "LinkNYC marks a radical step even for Google. It is an effort to establish a permanent presence across our city, block by block, and to extend its online model to the physical landscape we humans occupy on a daily basis...And every place that signs on will become another profit centre in Google's advertising business, even as it extends its near-monopoly on information about our online behaviour to include our behaviour in physical space as well".

By the establishment of Sidewalk Toronto, as a particular and tangible project, the level of cynicism has crucially risen. Woyke (2018) reported clearly as following "Sidewalk plans to let third parties access the data and technologies, just as developers can use Google's and Apple's software tools to craft apps. In fact, Sidewalk anticipates that 80 per cent of the work on Quayside will involve these third parties." The problem of the accountability of third parties' data usage became escalated and turned into a crisis in October 2018 with the resignation of the data privacy consultant Ann Cavoukian, a well-respected public figure with her 'Privacy by Design'¹⁷ framework. Ann Cavoukian was defending mainly the de-identification of collected data. She is insisting as follows; "From the beginning, I wanted it to be a smart

¹⁴ <http://senseable.mit.edu/>

¹⁵ <https://www.intersection.com/> and <https://www.link.nyc/>

¹⁶ <https://www.transportation.gov/>

¹⁷ Privacy by Design defined as "a framework that seeks to proactively embed privacy into the design specifications of information technologies, networked infrastructure and business practices, thereby achieving the strongest protection possible." at <https://www.ryerson.ca/pbdce/about/ann-cavoukian/>

city of privacy and not a smart city of surveillance.” CBC News reports that Cavoukian decided to resign after a meeting where; “Sidewalk Labs revealed that, while it has committed to stripping all of the data it collects of personal identifiers, it could not guarantee that other groups participating in the project would do the same.” (CBC News, 2018). Few weeks before Cavoukian Saaida Muzaffer, a member of an expert panel for Waterfront Toronto has resigned declaring Sidewalk Labs’ data privacy policy as “a blatant disregard for resident concerns about data and digital infrastructure”. (Vincent D. , 2018)

In line with Cavoukian, the CEO of Apple Tim Cook (Vincent J. , 2018) declares that the “data has been weaponized against users”. He concludes that the data privacy is one a fundamental human rights: “Rogue actors and even governments have taken advantage of user trust to deepen divisions, incite violence, and even undermine our shared sense of what is true and what is false. This crisis is real. It is not imagined, or exaggerated, or crazy”. Ann Cavoukian agrees with Tim Cook’s furious attitude, she considers Google is not following the data collection procedures correctly. She continues, with reference to Cook’s speech on a local television interview after her resignation, as follows: “Data minimization is all about de-identifying data...So you don't know whom the data is linked to. This is critical and this is increasingly the model I am hoping we will involve into.” Finally, she adds: “You cannot have free and open societies without a foundation of privacy”. (Cavoukian, 2018)

From the point of view of Sidewalk Labs, Aggarwala declares their data privacy policy as following; “number one, we're going to be completely transparent about the whole data collection; number two, collect data for the purposes of making the neighbourhood work better, not because we are trying to targeted advertising, or monetise it in some way.” (Aggarwala, 2018). What is missing from his speech again is the third parties. He continues at the same interview in June 2018; “We want to embrace this idea of privacy by design. We want to develop systems that will not capture personally identifiable information unless it is intended for the purposes of something where we are identifying an individual so that would be on a consensual basis.” Eventually after the crisis of trust in October Sidewalk Labs has issued an update¹⁸ for data privacy which has also been concluded in Draft Site Plan under the name of “A Made in Canada Approach”. The update was touching upon the points of data minimization, ethical usage of technology, and meaningful consent with a clear statement of “Commitment to not share or link personal data with 3rd parties - including other Alphabet companies (i.e. Google) without consent.” It may be cautiously interpreted as a victory for data privacy concerns, at least at the policy level.

The major anxiety beyond personal privacy, thus, is the provided opportunity for Google to run a city without being elected. Sadowski (2017) warns; “this model of creating our urban future is also an insidious way of handing more control – over people, places, policies – to profit-driven, power-hungry corporations.” Such a future may not be identified with democracy but an autocracy. What is more, recently Sidewalk Labs promised to help to the City of Toronto in various public services and investments that they think appropriate to promote the Quayside projects such as financing the Waterfront LRT, constructing more affordable housing as well as a tall timber factory to supply building material and finally helping for the infrastructure of the Innovation District. However, in return, they explicitly covet the revenues. In other words, they basically want to collect taxes as a private international company: “We would then recoup those investments through sharing in those revenue streams if and when they materialize.” as declared by Doctoroff (2019) .This final ‘update’ definitely got reactions, since the private value capture is calculated to be approximately \$6 billion over 30 years (Goldberg, 2019). *#BlockSidewalk* gains popularity among opponents¹⁹ mainly based on the common objection against sharing property taxes and development fees (Bliss, 2019). That is, perhaps, an unexpected dilemma at the field of democracy, governance and citizenship rights about translating ‘connective qualities’ of

¹⁸ <https://medium.com/sidewalk-talk/an-update-on-data-governance-for-sidewalk-toronto-d810245f10f7>

¹⁹ <https://www.blocksidewalk.ca/>

online platforms into ‘public values’. Google aspires to execute its well-functioning business plan to urban planning field, however, it threatens the democratic planning processes and eventually fundamental notions of urban public space such as a place of freedom, creativity, inclusion, opposition and solidarity. At this apparent deadlock regarding smart urban planning, there is an urgent necessity to redefine a more balanced platform society; “an open society that anchors its public values in a transparent and accountable manner.” (Van Dijck, Poell, & De Waal, 2018)

Findings from the Visual Analysis

For the visual analysis, I used three sets of visuals issued on different dates spread over three years from 2017 to 2019. First, Set 1 was issued by Sidewalk Toronto on 17 October 2017 under the name of *Vision Images*, sharing the conceptual visuals for the first time. CGI’s from the first set are sketchy and ambiguous at architectural decision-making, providing a general overview including visions for the community, digital infrastructure, housing, mixed-use, mobility, public realm, and sustainability. I have only excluded the digital infrastructure vision since it was a birds-eye perspective and focused on seven images that represent the future public space in Quayside Smart City. Set 2 was presented on 14 August 2018 during the third Roundtable meeting. This set contains three types of images; a site plan diagram, two photo-realistic CGI’s – one showing the ‘building raincoat’ feature and revealing the author as local Michael Green Architects²⁰ – and four sketchy visuals representing roughly the public life. I have eliminated the diagram and worked on the other six images. Finally, Set 3 was announced as a project update on 14 February 2019. The final set is comprising of one bird-eye perspective overlooking the layout in standard SketchUp wireframe output and six photo-realistic CGI’s, on which I have worked on. For the final set, two internationally-renowned architectural offices are credited; Heatherwick Studio from UK and Snøhetta²¹ from Norway. An image looking to the silhouette of Quayside from the lake and a second unspecified interior is credited with Snøhetta, while images representing a building exterior, campus, courtyard and innovative campus are credited with Heatherwick Studio. All the images in this set, apparently, are coherent and reflect the same feel or atmosphere regardless of the architectural practice it is associated with. What is more, exceptionally, I couldn’t find any information nor public declaration available by architects related to the Quayside project. Sidewalk Toronto, as the developer of the project, is much more at the foreground compared to the architects.

I should also note that the software used to draw the sketchy images for first two sets of visuals was SketchUp, an easy-to-use, largely available and mostly free online tool for 3D modelling and design owned by Trimble²². However, SketchUp was owned by Google until 2012 yet the software is still working highly compatible with other Google applications and tools. For the rest of the images, I could not identify the software.

²⁰ <http://mg-architecture.ca/>

²¹ <https://snohetta.com/>

²² <http://investor.trimble.com/news-releases/news-release-details/trimble-enhance-its-office-field-platform-acquisition-googles?releaseid=667690>

Sidewalk Toronto as Utopia

After briefly presenting the Quayside project and the related criticism, I will unfold my findings from the visual analysis that I have conducted. Before that, I will mention a particular article that has followed similar paths as I did. Breaking apart common critical approaches, Molly Sauter (2019) considers Sidewalk Labs Quayside project as a visual representation of a utopia; “artful invocations of phantasmic images that are intended for and accessible to an audience outside the geographic boundaries of the project proposed”. What is more, Sauter draws parallels between Quayside visuals with colonial accounts to persuade more to immigrate this time to the digitalized Smart Cities.

Her review is based on two sets of visual representations that were available on the official web site. The third set of visuals with more realistic CGI’s has been issued and architectural practices have been revealed approximately two weeks after Sauter’s article has been released. Thus, it should be noted that her critic does not cover , so far up-to-date, the entire process of the presentation. However, she pinpoints several aspects that deepen my argument. First, she asserts the significance of the image for a future promise, yet admits what is only available now is the sales documents in visual form: “Sidewalk Toronto is, still, a digitally-bounded project, built out of renderings with persuasive power but no planning utility. Lush, colorful, calming, and detailed to a fantastic degree, these illustrations are a major component of the Sidewalk sales pitch.” Second, she notices that the municipal and manual labor or services are ‘automated away’ in visuals. Finally, she argues that visuals (at least for the first two sets) are explicitly intended not to be binding. Even though they are abundantly circulated, Sidewalk Labs insist that they are not indicative plans for the design.

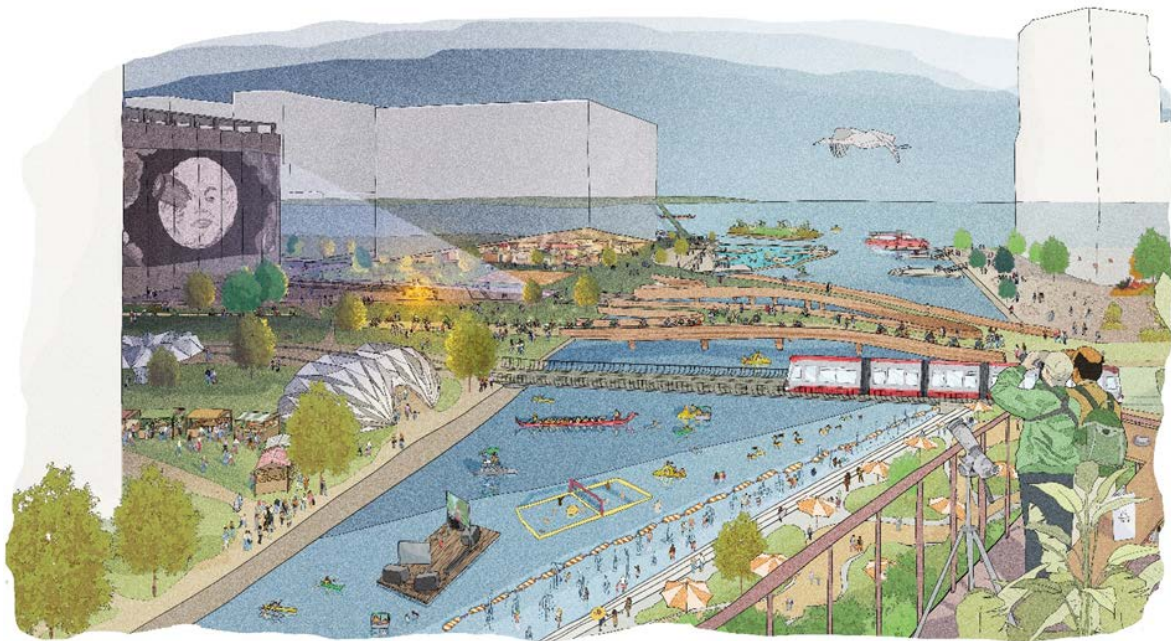


Figure 8 The Silo Park By Sidewalk Toronto from Set 2

Karim Khalifa, the director of Building Innovation at Sidewalk Labs, names Andrew Edwards a former ‘intern’ from Heatherwick Studio²³ whom Sauters failed to reach after several attempts. Sauters quotes from Khalifa strikingly as following: “I just tell him a story and he goes,” said Kalifa, who, just an hour before, had given a presentation on the adaptability of the buildings planned the Sidewalk Toronto site, illustrated almost entirely with Edwards’s work.” In the article, Sauter deliberately urges upon an image

²³ <http://www.heatherwick.com/>

that she thinks has synoptic significance (Figure 8); “a particularly stunning image took a sweeping view of what Sidewalk called ‘Silo Park’”. Sauters interprets that image illustrated by Edwards as following: “Edwards here shows the Silos illuminated by an iconic image from Georges Méliès’s 1902 silent film *Le Voyage Dans la Lune*. The viewer is positioned quite close to an elderly couple, looking over the scene below through binoculars as if from a designated scenic overlook. Edwards has, through his drawings, rendered us all welcome tourists in this brief glimpse of the future.” (Sauter, 2019)

Interpreting the Visuals from Quayside

Content Analysis - Coding Visuals

First of all, I began by applying content analysis through coding categories afore-mentioned by using the software titled Nvivo. All in all, categories used to make the visual analysis sought to be understood as possible intentions or hints related to the public life imagined to happen in Smart City scenery of Quayside Toronto. I concluded coding in two main categories with sub-categories; ‘Twelve Quality Criteria’ with 38 sub-categories and ‘Incomplete Form’ with 50 sub-categories. I used software to code and make a comparative frequency count due to different sets.

Performing content analysis by coding categories, I have focused on some salient points that are illustrated below represented as corresponding shades of blue (Figures 9-11). First I have applied ‘Incomplete Form’ from Sennett (2017) through categories that I have personally developed according to his article titled *The Public Realm*. Below main coding categories are represented with counts, respectively as; spectator architecture, resilience, porosity and cosmopolitanism. The darkest shade is associated with the highest counted code category while the faintest shade has zero counts²⁴. It is certain from the analysis that the spectator architecture, focusing on assorted public activities, got more counts in Set 1, whereas cosmopolitanism, investigating the diversity of the public, is the most noticeable focal point in Set 2. However, in the final set of visuals, all of the categories reach into a more balanced state.

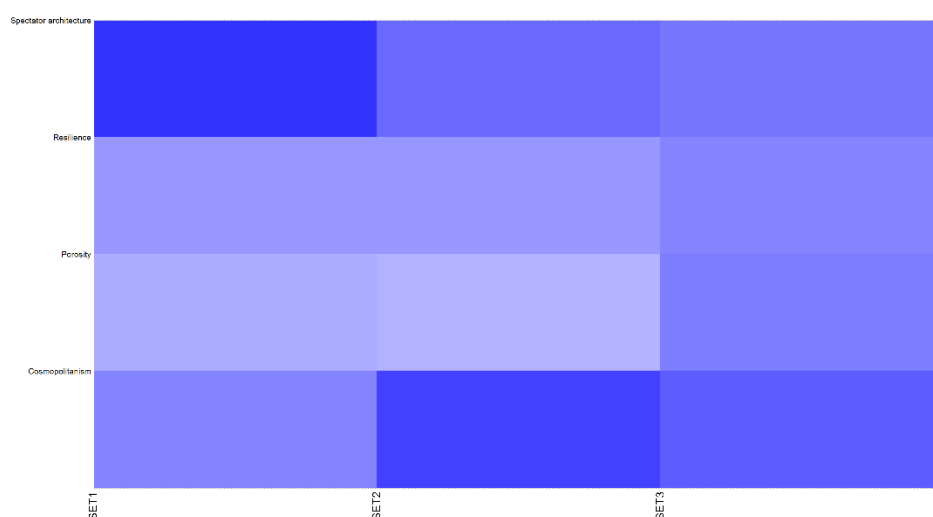


Figure 9 The Incomplete Form with main categories

For the second round (Figure 10), I have analyzed the visuals through sub-categories and focused on the most apparent categories with counts over the average of 8 out of 16. For the ‘Incomplete Form’, most apparent attributes in Set 1 are a bike, eating-out, trees on the ground, birds, ground floor closed (under the category of porosity), different ethnic groups and ages. Bikes have the exact same count (10)

²⁴ Values in parenthesis represents how many times a coding category is counted in this chapter.

in each set of visuals. It is obvious that biking is promoted consistently both as a sports activity and a means of sustainable transportation. I have listed 'eating-out' as a shopping activity. In Set 1 it is used amply to illustrate a vibrant public realm, however in other sets it seems to deescalate (from 9 to 3). 'Trees on the ground' stands out to the most counted category (14) in the first set, listed under the resilience main category. While losing relatively its significance in sets 2 and 3, 'trees or shrubs on the buildings' became more noticeable (from 4 in Set 1 to 11 in Set 3). The ground floor permeability of building masses is less significant in the first set but it accelerates in subsequent sets. This is due to the conceptual nature of first drawings evolving from sketches to a state that is more definitive on the architectural form, material and partitions. Different age groups and ethnic backgrounds represented at an average level (8) in the first set. However, images related to ethnic diversity reaches at their peak in Set 2 (15), then dramatically descends again in Set 3 (6). This may be correlated with the content of the images; Set 2 represents more roughly the public realm, whereas Set 3 illustrates more concretely semi-public exteriors and interiors with commercial and residential functions. Definitely, it is remarkable that ethnic minorities are represented less in such built environments and public spaces. On contrary, the representation of different age groups is perpetually increasing in subsequent sets (from 8 to 12 and 16), reaching the most counted category in this chart.

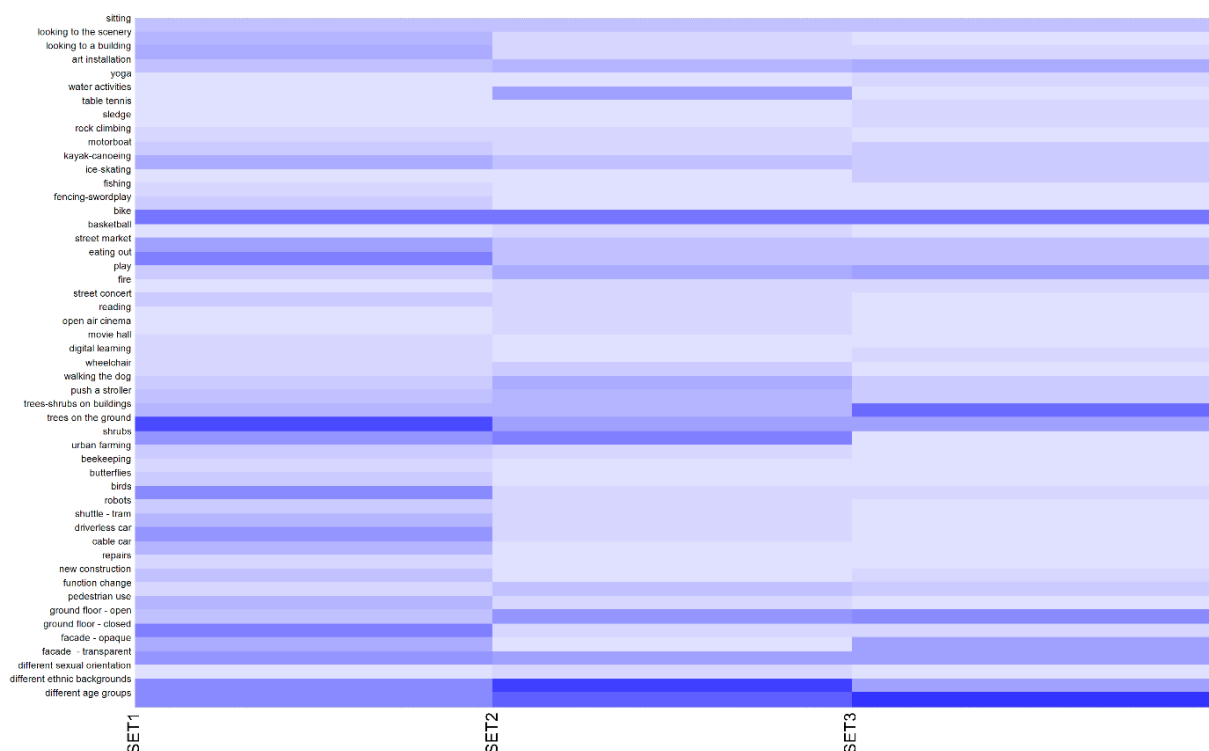


Figure 10 The Incomplete Form with sub-categories

They are also other intriguing aspects other than the most remarkable ones. For instance, the play category became more apparent in later sets or visibility of different sexual orientation found a place only once in Set 2. In general, there are a broader array of activities in the first set, accompanying activities such as pushing a stroller or walking the dog are quite common. However, the most assorted category is sports where I counted twelve different sports activity ranging from fencing and ice-skating to yoga and kayak/canoeing. This variety diminishes slightly in number in Set 3, yet the variety remains almost the same. What is especially remarkable is the category of 'adaptive building feature' under the main category of resilience. Robotics, driverless vehicles and others are often common for visuals in Set 1, however, they disappear in the final sets. It can be deduced that the experimental character of the visuals are diminishing whereas the impressive atmosphere becoming more imminent.

For the second category, I have applied ‘Twelve Quality Criteria’ from Jan Gehl (2010), whereas I have used categories exactly as they are presented by Gehl with main categories dedicated to comfort, delight and protection. Most noticeably the intensity on comfort in Set 1 diminishes in sets 2 and 3 as shown in the chart below (Figure 11). Nonetheless, I should admit that I found it difficult to interpret some categories based on subjective evaluation such as fine view, good design and detailing, more tangible attributes such as good materials and good surfaces, intangible aspects such as breeze and noise levels or attributes depending on time such as overlaps, lighting, day and night use. Gehl’s criteria stand out first to inquire existing public spaces but also secondarily to shade light over the future ones. Hence I consider the ‘Twelve Quality Criteria’ as an important tool for making the comparison.

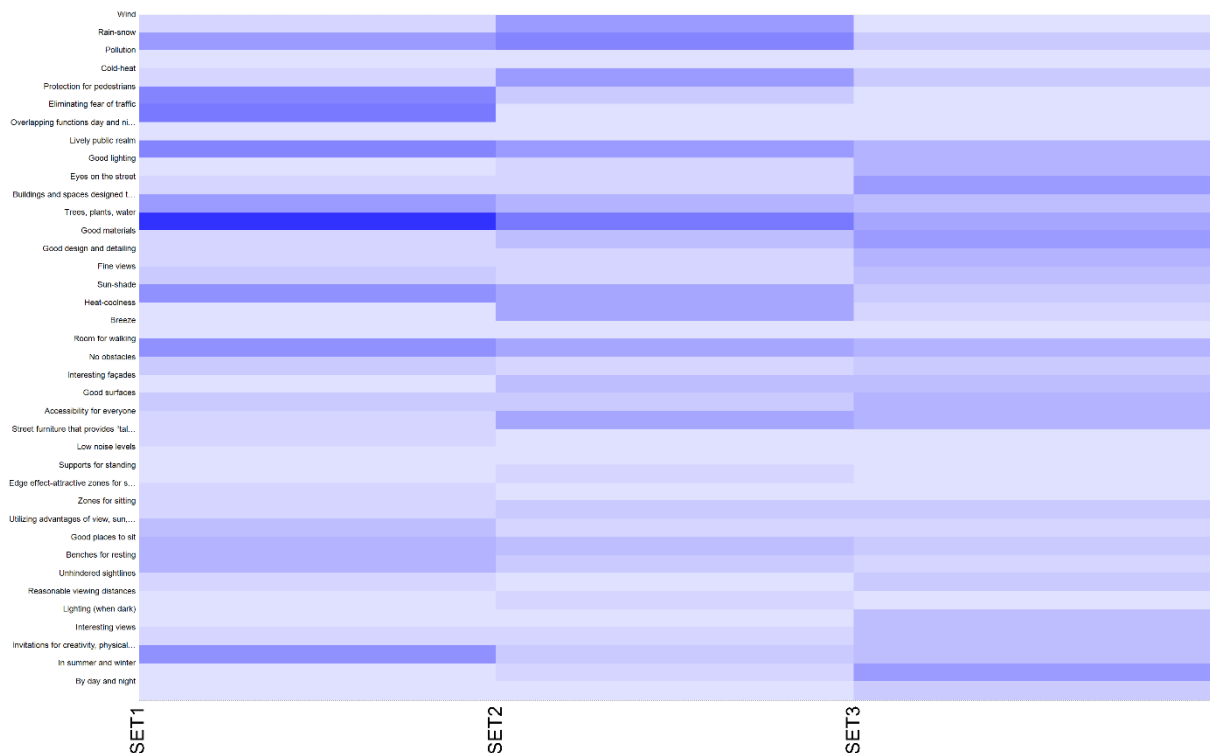


Figure 11 The Twelve Quality Criteria through sub-categories

Applying the ‘Twelve Quality Criteria’ I have evaluated categories which again had more than average counts (above 7 over 15). Most apparent attributes in Set 1 is the ‘trees, plants and water’ sub-category under the delight main category as positive sensory experiences. However, the emphasis on natural entities decreases in subsequent sets (from 15 in Set 1 to 5 in Set 3). Categories related to the protection against traffic and accidents hold a significant part, especially in Set 1 linked with lively public realm and room for walking. However, the protection or feeling safe at the street level is strongly related to the presence of driverless vehicles, public transportation and shared car schemes, thus, the strong emphasis on this category diminishes sharply in sets 2 and 3. Another apparent point in Set 1 is the prevalence of invitations for creativity, physical activity, exercise and play listed under the comfort category. In Set 2 protection against unpleasant sensory experiences are highlighted, even so, there aren’t coming forward particularly in sets 2 and 3. Categories that have been counted the least are the ones associated with pre-mentioned hardships of subjectivity, tangibility and temporariness. Nevertheless, it seems there are some categories that were hard to relate with any visuals apart from those hardships such as, for instance, opportunities for standing or *talkscapes* and noise levels.

Semiological Analysis and Compositional Interpretations

I have so far conducted a rather analytical analysis while evaluating the sets of visuals. To conclude, I have used two main categories to question what is shown and what is not through the images. Furthermore, I managed to detect some alterations in sequential sets. Rose (2016) asserts that content analysis is concentrating only on the image itself whereas other sites of modalities such as audiencing and production of the image remain largely untouched. Thus, I will develop a more subjective and profound one using semiology and compositional analysis. I will first start with some remarks through semiological concerns looking at the 'how of representation'. Accordingly, I will attempt to look at images not in parts but as a whole, assessing the compositional characteristics. In particular, I want to initiate elaborating from the outcomes of content analysis with associated images.

The opening image of the final set issued in February 2019 is at below (Figure 12). The image places two kayaks at closest to the viewer – one of many sports activities illustrated through visuals of the project. The horizon line is low, implying a view from the eye level of the audience. In other words, the image posits the viewer at an imaginary kayak, offering a supposedly stunning look to Quayside from Lake Ontario - as if the viewer is one who can afford to kayak. Melcher (2014) astonishes the abundance of kayaks in CGI's: "Renderings have a special affinity with kayaks; everybody is kayaking all the time. It's, apparently, how these folks get around—paddling between 'Market', 'Pharmacy' and 'Residential'". He also notices the risen popularity of fireworks, birds, butterflies, hot-air balloons and drones to display at CGI's.



Figure 12 Building Exterior by Snøhetta from Set 3

The CGI represents a dramatic atmosphere at sunset since the lower parts of the building are well lit and the waterfront looks crowded. In fact, the location of the sun at the sunset is wrong since the orientation of building complex is actually towards the south-east direction. Therefore, the illuminated part of the image should be the right side, if it was put geographically correct. Nevertheless, the sunset provides a dramatic background in blue and pink shades, where the project shines with the last beams from the sun. The view from the lake contributes to the depth of the image with reflections. To emphasize the project, even more, the surrounding buildings left as white masses without material information, except the existing Silo Dam building at the rightest. Even though realistically shown, trees are spreading from the Quayside project to the skirts of surrounding buildings, adding to the ambiguous

feel of the image. Quayside complex is not situated at the exact centre of the image, it is tilted a bit to the left. This composition gives a large place at left for the kayaker and lets the viewer better engage with a future scenario.



Figure 13 Current state of the Quayside provided by Google Earth

Most of the scenes are intensely crowded with happy people. As Melcher (2014) argues Sidewalk Toronto wants to ensure the viewer that the Quayside is going to an ideal smart city: “There are crowds, but nothing is crowded. There are cars, but there is no traffic. The scalies²⁵ dropped into these polished scenes are as happy as can be—why wouldn’t they be?” Another aspect is related to the diversity of the crowd. As stated earlier in first sets, especially in Set 1, human figures have been displayed more diversely as a part of the lively public realm. For instance, in Figure 14 people at the focal point of two-point sketchy perspectives have rather darker skin colours with self-confident postures. However, in Set 3 visuals are rather representing semi-private commercial interiors, not just open public spaces as it was in Set 1 and remarkably the crowd from the final set is almost homogeneous. For instance, in Figure 15 out of sixty-eight people displayed, only two are likely to be associated with a diverse ethnic background. What is more, they are not at central positions nor posed in a confident manner. I can argue that commercial environments aim to sell a safe atmosphere for shopping with less ethnical heterogeneity.



Figure 14 Public Realm Vision from Set 1

²⁵ Scalies is used for the human figures at CGI's.



Figure 15 Interior by Snøhetta from Set 3



Figure 16 Details from Building Interior

By the same token, the display of people with different sexual orientation is noticeably few. I could only detect one image in Set 2 with a probably-gay couple. It seems that the representation of the diversity is a subject to consider for Sidewalk Toronto which they need not overstate. It brings to mind Benetton's bold stance while putting publicly controversial people or subjects at the focal point of the advertisement. Comparatively, the Sidewalk Toronto's approach is quite discreet. King (2015) points out the unreal yet sterile climate of the CGI's: "The city imagined by designers and developers is a monoculture of the wired and young", as he continues there is no room for common 'nuisances': "Panhandlers are never in the background" nor "Graffiti does not scar the walls".



Figure 17 Exterior by Michael Green Architects Set 2



Figure 18 Detail

The magical feel of the image climaxes chiefly at the final set. In Set 3 nearly all images are, colloquially, sugar-coated. Figure 19 is representing a pleasantly lid snowy evening where children play and ice-skate. Lush vegetation is available at wooden balconies, yet they are green behind the transparent glazing. The image brings to mind Pieter Bruegel's²⁶ vivid and intense winter pictures with colours and composition. The lively public realm shown in harmony here is important since the year-long use of outdoor public space is widely promoted by Sidewalk Labs. Thus, the image aims to illustrate a fairy-tale style scenery, where heated walking surfaces and 'building raincoats' are used to enable public use even during unpleasant weather conditions. The level of the horizon is again very low suggesting an eye-level look to the 'magical' atmosphere with lights. It is the kind of magic that resonates with Williamson (1978).



Figure 19 Building Exterior Picture Plane for Heatherwick Studio

Sidewalk Labs are seemingly committed to concretize the miracle – at least partly. In order to achieve longer use of outdoors several architectural and design offices collaborate and produce exact scale not-so-fairy mock-ups²⁷. So-called 'building raincoats' are deliberately covering streetscapes, which are called as 'stoas', an explicit reference to the Greek Stoas – a covered walkway for public use. At the presentation where Set 2 of visuals has been issued, Sidewalk Toronto (Roundtable Three, 2018, s. 52) defines stoa as following: "Sidewalk Toronto's stoa is a new way of thinking about ground floor space that uses a highly flexible, yet standardized system to support retail, production, and community uses. In response to the community's changing needs over time, stoa can evolve and adapt to support a variety of different uses." The definition is coupled with a photograph from the restored Stoa of Attalos in Athens and the reproduction of the famous fresco of *The School of Athens* by Renaissance artist Rafaello (both uncited at the report). Berger (1972) notes the analogy between publicity and classical cultural references by emphasizing the propagation of self-belief: "Publicity is, in essence, nostalgic. It has to sell the past to the future. It cannot itself supply the standards of its own claims. And so all its references to quality are bound to be retrospective and traditional. It would lack both confidence and credibility if it used a strictly contemporary language." But the miraculous point is to render a place-bound urban form – in this case, the Mediterranean stoa – available for all, regardless of the location by virtue of smart building textures.

²⁶ Hunters in the Snow (1565) by Pieter Bruegel the Elder

²⁷ <https://www.archdaily.com/913095/sidewalk-labs-tests-a-raincoat-for-buildings-in-toronto>



Figure 20 The Stoa through changing seasons by Sidewalk Toronto

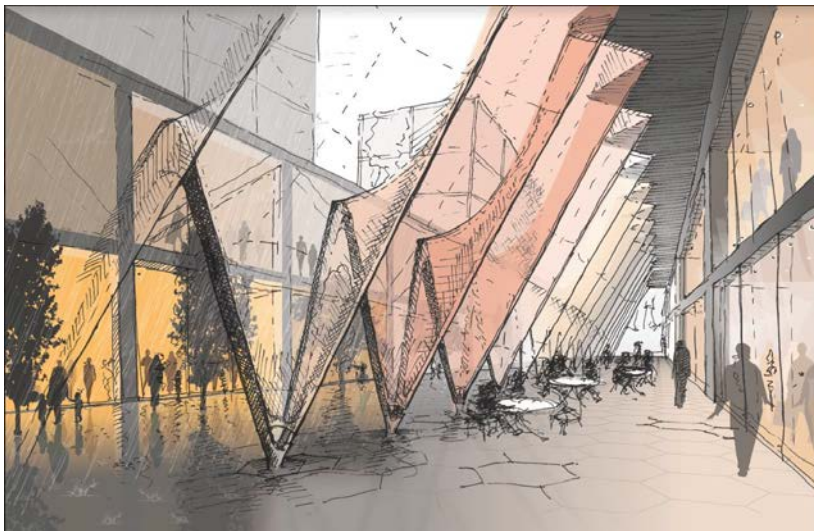


Figure 21 Building raincoat with hexagonal paving tiles by Partisans



Figure 22 Mock-up by Sidewalk Labs

Conclusion and Discussion

The recent emergence and prevalence of Smart Cities discourse indicate the fact that the urban environment will be more and more shaped by digitally assessed optimization and control. On one hand, we are witnessing dazzling urban innovations that offer a more efficient, sustainable, full, smart way of living. On the other hand, the constant data traction, computability and digitalisation bother us concerning our fundamental right of privacy, inclusion and democracy. Hollands (2015) asserts the 'Right to Use Technology' by referring Harvey's famous 'Right to the City'; "right to shape the city using human initiative and technology for social purposes to make our cities better and more sustainable." This approach reclaims technological development but prioritizes human will, creativity and agency. Accordingly, each and every category, including the design of urban public space, should stipulate the experimental and democratic character while coming out against the possible tyranny of capitalism and neoliberal policies.

The aim of this study was to reveal how Smart Cities are being imagined and designed. Smart City is still a discursive category that is largely associated with the future and, as I have argued before, it is not yet possible to experience entirely a Smart City. Therefore, I have chosen to decipher how it is represented architecturally and conducted a comprehensive visual analysis to investigate how public spaces of the Quayside project are conceived by Sidewalk Toronto. Hereby, I expected to find clues or tacit implications to designate how future Smart Cities are conceived literally by Google – the ubiquitous

technology giant as well as one of the biggest provider of the platform economy. I have briefly presented the Quayside urban development project followed with the criticism against it. Later, I have unfolded my visual analysis in two main parts over three sequential sets of visuals: First the analytical part with content analysis by coding categories and, then, second with semiological and compositional analysis of the salient outcomes from the content analysis. I assert that the categories I have applied were pointing out well-functioning public spaces and they functioned exhaustively and coherently in order to evaluate the Quayside project which has been represented as a Smart City. However, I should admit that coding categories were not entirely exclusive through overlaps as well as holding a level of ambiguity, especially considering Gehl's 'Twelve Quality Criteria' categories that were at times intangible or subjective.

I can briefly conclude that the overall experimental character correlated with diverse public realm at first sets remarkably diminishes at the final set. This includes the diversity of promoted activities, heterogeneous crowd and alternative means of transportation and logistics. However, it can be easily observed that the visual quality of the images has risen to a great degree. This increase also indicates the emphasis of the persuasive effect with reoccurring referents of nature and magic. Nevertheless, at the final set, visuals from the project accommodate hardly any progressive nor innovative sign of a smart urban future. That excludes also afore-mentioned ideas or innovations on cities such as robot delivery, driverless cars, innovative waste management ideas or adaptive traffic lighting. However, the features that have been highlighted the most are widespread use of timber as the main building material and the year-long usage of public spaces by virtue of adjustable building raincoats and pavement tiles.

Although Smart City stands out to be a powerful and attractive concept for the majority of stakeholders concerning the cities, I have initiated this research with the knowledge that there is also serious and understandable opposition. By and large, this study's focal point was Google as a new player at the urban design, planning and development scene. At first, I had underestimated how daring Google's ventures might be. However, besides the common allurements for better functioning and optimized urban environments, how Smart Cities are being conceived and imagined to be executed is threatening the way democratic urban spaces are designed, used and maintained. The research on Sidewalk Labs unveils that, especially after Sidewalk's disclosure on the desire for public taxes in return of proposed investment, this new player precariously candidates to rule and make elected local governances and planning instruments functionless. I would argue that this intention has also been proven through my visual analysis of the Quayside project showing the switch both in content and visual qualities.

To conclude, I want to emphasise what is previously stated; how we imagine the future is the commencement of its construction. Largely circulated urban image-making for future Smart Cities are part of the broad notion of imagination incorporating hopes, dreams, ambitions, aspirations, urges, expectations, insights or cosmologies. All technological developments, including the digitalization, have been accomplished through the agency of the human mind and imagination and therefore are within the scope of causality and human responsibility. Kenney & Zysman (2016) rightfully pinpoint that: "The technologies—the cloud, big data, algorithms, and platforms—will not dictate our future. How we deploy and use these technologies will." Instead of technology giants, that are already monopolizing the everyday life, at the age of the digital revolution it is us who are in charge of dreaming a better future and we have to ask now questions about how the future might be, what is a truly Smart City or how public space in Smart Cities should function. Smart Cities doesn't necessarily be advertised as stereotypes nor "setting(s) for the fulfilment of publicity's promise of the good life" as Berger (1972) puts. Yet they shall be conceived as urban spaces to re-think our relationship with the environment, metabolisms, modes of production and democracy. We don't need a perfect world with an over-determined form that is already imagined for us, we, as people, have to imagine, design and build smart, open and incomplete urban forms of tomorrow.

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Appendices

Coding Categories

1. Richard Sennett's Incomplete Form

- Resilience
 - Adaptive building features
 - Function change
 - New construction
 - Repairs
 - Robotics
 - Driverless vehicles
 - Cable car
 - Driverless car
 - Shuttle – tram
 - Robots
 - Environmental vitality
 - Animals
 - Birds
 - Butterflies
 - Natural entity
 - Shrubs
 - Trees on the ground
 - Trees-shrubs on the buildings
 - Farming
 - Beekeeping
 - Urban farming
- Porosity
 - Ground floor permeability
 - Closed
 - Open
 - Facade material
 - Transparent
 - Opaque
 - Pedestrian use

- Spectator architecture
 - Interaction
 - Accompanying
 - Push a stroller
 - Walking a dog
 - Wheelchair
 - Cultural
 - Digital learning
 - Movie Hall
 - Open-air cinema
 - Reading
 - Street concert
 - Fire
 - Play
 - Shopping
 - Eating-out
 - Street market
 - Sport activities
 - Basketball
 - Bike
 - Fencing-swordplay
 - Fishing
 - Ice-skating
 - Kayak-canoeing
 - Motorboat
 - Rock climbing
 - Sledge
 - Table tennis
 - Water activities
 - Passive activities
 - Art installation
 - Looking somewhere
 - To a building
 - To the scenery
 - Sitting

- Cosmopolitanism
 - Different age groups
 - Different ethnic backgrounds
 - Different sexual orientation

2. Jan Gehl's Twelve Quality Criteria

- Protection
 - Protection against traffic and accidents— Feeling safe
 - Protection for pedestrians
 - Eliminating fear of traffic
 - Protection against crime and violence — Feeling secure
 - Lively public realm
 - Eyes on the street
 - Overlapping functions day and night
 - Good lighting
 - Protection against unpleasant sensory experiences
 - Wind
 - Rain/snow
 - Cold/heat
 - Pollution
 - Dust, noise, glare
- Comfort
 - Opportunities to walk
 - Room for walking
 - No obstacles
 - Good surfaces
 - Accessibility for everyone
 - Interesting façades
 - Opportunities to stand/stay
 - Edge effect/attractive zones for standing/staying
 - Supports for standing
 - Opportunities to sit
 - Zones for sitting
 - Utilizing advantages: view, sun, people
 - Good places to sit

- Benches for resting
- Opportunities to see
 - Reasonable viewing distances
 - Unhindered sightlines
 - Interesting views
 - Lighting (when dark)
- Opportunities to talk and listen
 - Low noise levels
 - Street furniture that provides “talkscapes”
- Opportunities for play and exercise
 - Invitations for creativity, physical activity, exercise and play
 - By day and night
 - In summer and winter
- Delight
 - Scale
 - Buildings and spaces designed to human scale
 - Opportunities to enjoy the positive aspects of climate
 - Sun/shade
 - Heat/coolness
 - Breeze
 - Positive sensory experiences
 - Good design and detailing
 - Good materials
 - Fine views
 - Trees, plants, water